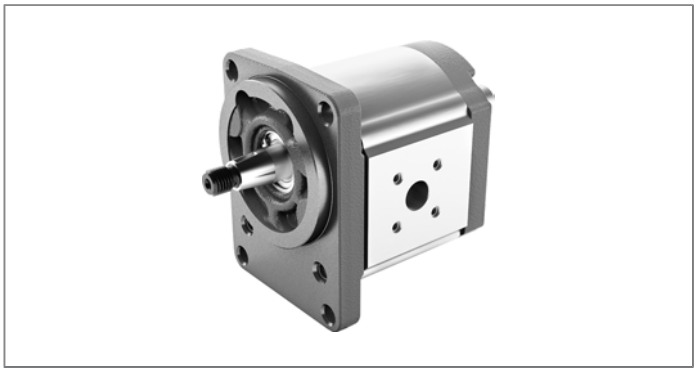
External gear pump Standard performance
AZPW

RE 10090
Edition: 06.2016



- Platform F
- Fixed displacement
- Sizes 4 to 22
- Continuous pressure up to 190 bar
- Intermittent pressure up to 210 bar

Features

- Consistently high quality due to high-volume series production
- Long service life
- Slide bearings for high loads
- Drive shafts conform to ISO or SAE and customer-specific solutions
- Line connections: Connection flange or screw-in thread

Contents

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Functional description

General

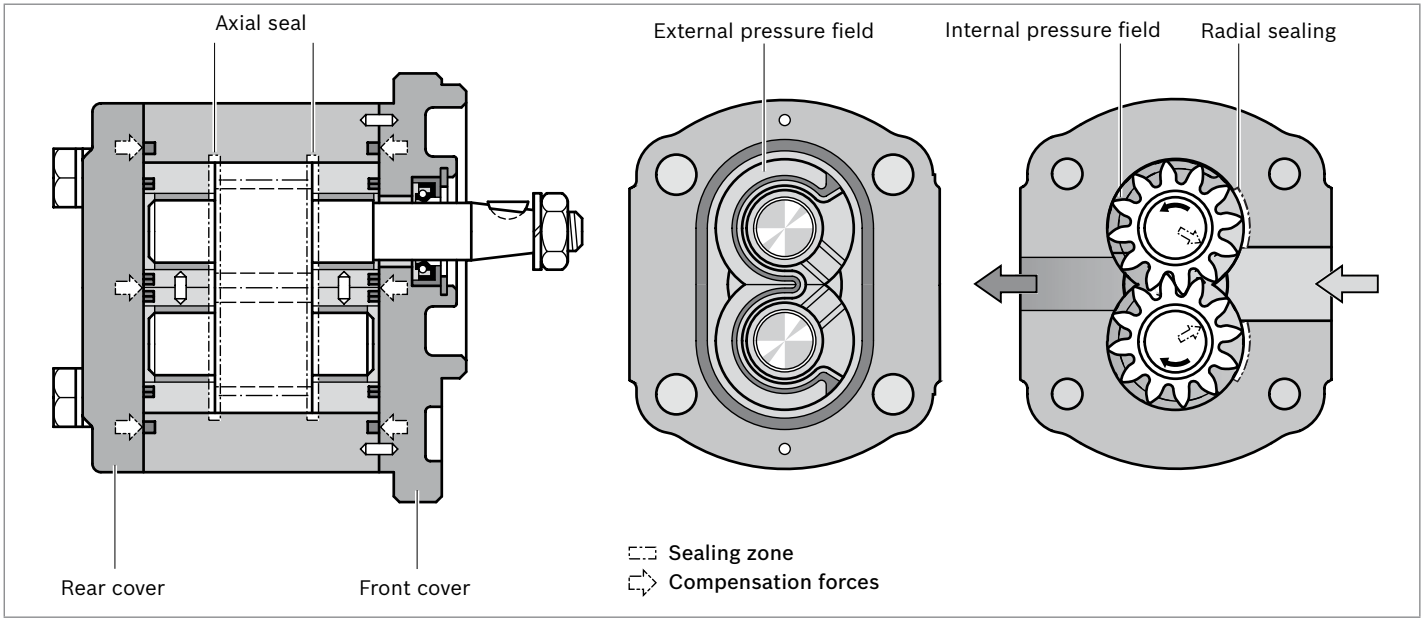
Rexroth external gear pumps are available as standard performance gear pumps in the series W. The displacements are graded by different gear widths. Further configuration variants are given by different flanges, shafts, valve arrangements and multiple pump combinations.

Construction

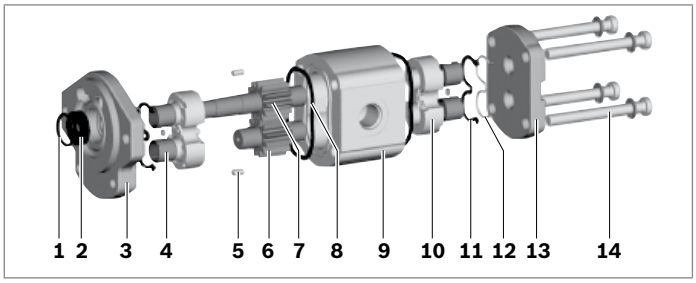
The external gear pump consists essentially of a pair of gear wheels supported in bearing bushes and the housing with a front and a rear cover. The drive shaft protrudes from the front cover where it is usually sealed by the shaft seal. The bearing forces are absorbed by slide bearings.

These were designed for high pressures and have excellent emergency running properties, especially at low rotational speeds. The gear wheels have 12 teeth. This keeps both flow pulsation and noise emission to a minimum. The internal sealing of the pressure chambers is achieved by delivery pressure-dependent forces. This ensures optimum efficiency. On the rear side, the movable bearing bushes are pressurized with working pressure and pressed as seals against the gear wheels. The pressurized compression springs are limited by special seals. The seal on the area between the gear teeth and the housing is ensured by the smallest of gaps that adjust depending on the pressure between the gear teeth and housing.

▼ **Axial compensation standard performance**

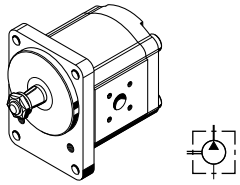
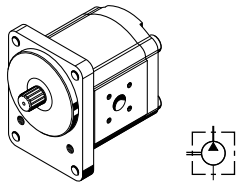
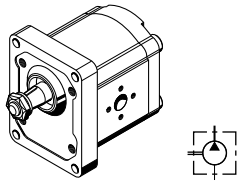
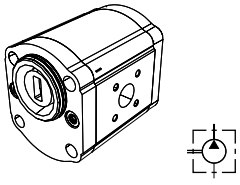
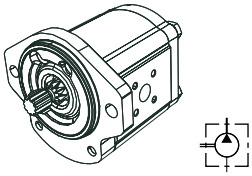


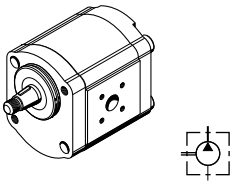
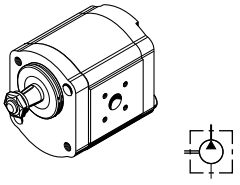
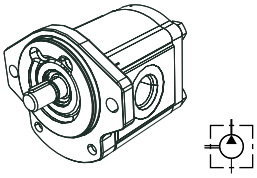
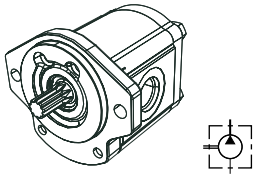
▼ **Layout AZPW** (schematic diagram)

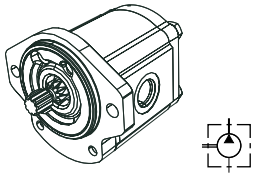
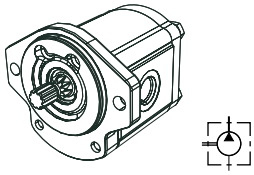
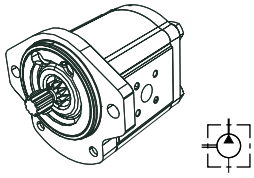
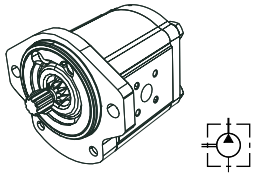


1 Snap ring
2 Shaft seal
3 Front cover
4 Slide bearings
5 Centering pin
6 Gear wheel
7 Drive shaft
8 Housing seal ring
9 Pump housing
10 Bearing bushing
11 Axial field seal
12 Support element
13 Rear cover
14 Hexagon screw

Product overview AZPW preferred types

Version	Page
	16
	17
	18
	19
	28
Line connections in Non-standard version	

Version	Page
	20
	21
	22
	23

Version	Page
	24
Suction port in the front cover	
	25
	26
Suction port in the front cover	
	27

Type code single pump

01	02		03	04		05	06	07	08	09	10	11		12
AZP	**W**	**–**		**1**	**–**							**B**	**–**	

External gear unit

01	External gear pump	**AZP**

Series

02	Standard performance, platform F	**W**

Series

03	Standard housing	**1**
	Stronger housing	**2**

Version

04	Phosphated	**1**

Size (NG)

05	Geometric displacement V_g [cm³], see "Technical data" on page 6	**004**	**005**	**008**	**011**	**014**	**016**	**019**	**022**

Direction of rotation

06	Viewed on drive shaft	clockwise	**R**
		counter-clockwise	**L**

Drive shaft			**Suitable front cover**	
07	Tapered shaft	1 : 5	**B, P**	**C**
		1 : 8	**O**	**H**
	Dihedral, claw		**T**	**N**
	Splined shaft	SAE J744 16-4 9T	**R**	**R**
		DIN 5482 B17 × 14	**B, P**	**F**
	Parallel keyed shaft	SAE J744 16-1 A	**R**	**Q**

Front cover

08	Rectangular flange	Ø80 mm		**B**
		Ø36.47 mm		**O**
	2-bolt flange	Ø82.55 mm	SAE J744 82-2 A	**R**
		Ø50 mm	connection variant 1	**N**
		Ø50 mm	connection variant 2	**P**
	4-bolt mounting	Ø52 mm	with O-ring	**T**

Line connection

09	Thread UNF-2B, SAE O-ring		**12**
	Square flange		**20**
	Square flange		**30**
	Non-standard size		**XX**

Sealing material

10	NBR (nitrile rubber)	**M**
	FKM (fluoroelastomer)	**P**

01	02		03	04		05	06	07	08	09	10	11		12
AZP	**W**	**–**		**1**	**–**							**B**	**–**	

Rear cover

11	Without valve (standard)	**B**

Special version

12	Serial number, e.g., S0001	**SXXXX**
	Without nut and washer	**S0007**
	For mounting on an axial piston unit and suction port in the front cover	**S0036**
	For mounting on an axial piston unit	**S0081**
	For mounting on an axial piston unit and line connections in Non-standard version	**S0593**

Note

- Not all of the variants according to the type code are possible.
- Please select the desired pump with the help of the selection table (preferred types) or after consultation with Bosch Rexroth.
- Further options are available on request.

Technical data

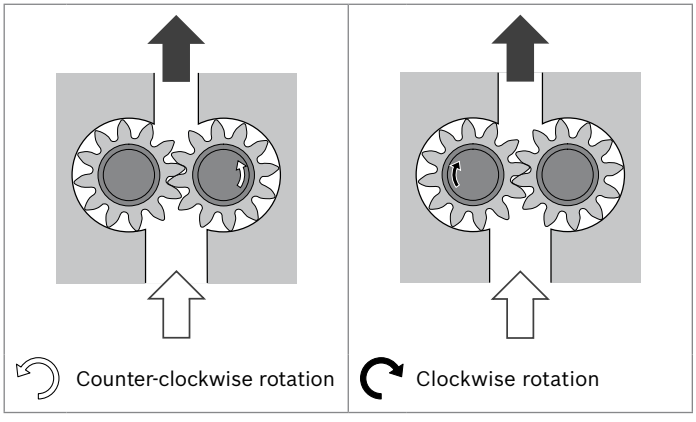
General				
Installation position				Any
Mounting type				See proposal drawing
Line connections				See proposal drawing The screw threads for the suction and pressure port, that are manufactured true to gauge, are deformed slightly due to the pump assembly. The mounting of conventional screw fittings in accordance with DIN EN ISO 1179, 6149, 9974 etc. is easily possible.
Direction of rotation (viewed on drive shaft)				Clockwise or counter-clockwise, the pump may only be driven in the direction indicated
Drive shaft loading				Radial and axial forces only after consultation
Ambient temperature range θ			°C	-30 to +80 with NBR seals -20 to +110 with FKM seals

Hydraulic				
Hydraulic fluid				Mineral oil according to DIN 51524, 1-3, however under higher load at least HLP compliant with DIN 51524 Part 2 is recommended. Please observe data sheet 90220. Other hydraulic fluids on request
Hydraulic fluid temperature range		θ	°C	-30 to +80 with NBR seals (NBR = nitrile rubber) -20 to +110 with FKM seals (FKM = fluoroelastomer)
Viscosity range	Permissible in continuous operation	ν	mm^2/s	12 to 800
	Recommended in continuous operation	ν_{opt}	mm^2/s	20 to 100
	Permissible for cold start	ν_{max}	mm^2/s	≤ 2000
Maximum admissible degree of contamination of the hydraulic fluid Cleanliness level according to ISO 4406 (c)				20/18/15[1)]

Note

- Safety requirements pertaining to the whole system are to be observed.
- Please contact us for applications with frequent load changes.
- The dimensions represent pumps for clockwise rotation. The position of the drive shaft or suction and pressure port changes for counter-clockwise rotation.

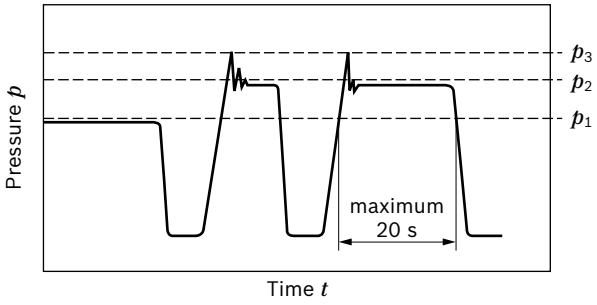
▼ **Direction of rotation, viewed on drive shaft**



1) For hydraulic systems or devices with function-related, critical failure effects, such as steering and brake valves, the type of filtration selected must be adapted to the sensitivity of these devices.

AZPW-1x				NG	004	005	008	011	014	016	019	022
Displacement, geometric, per revolution			V_g	cm³	4.0	5.5	8.0	11.0	14.0	16.0	19.0	22.5
Pressure in suction port		absolute	p_e	bar	0.7 to 3							
Maximum continuous pressure			p_1	bar	190	190	190	190	190	190	180	170
Maximum intermittent pressure			p_2	bar	210	210	210	210	210	210	200	190
Maximum pressure peak			p_3	bar	230	230	230	230	230	230	220	210
Minimum speed	at 12 mm²/s and	< 100 bar	n_{min}	rpm	700	700	700	700	700	700	700	700
		100 to 180 bar		rpm	1200	1200	1000	1000	800	800	800	800
		180 bar to p_2		rpm	1400	1400	1400	1200	1000	1000	1000	1000
	at 25 mm²/s and	p_2		rpm	700	700	700	700	700	700	700	700
Maximum speed	at p_2		n_{max}	rpm	3500	3500	3500	3500	3000	3000	3000	2500

▼ **Pressure definition**

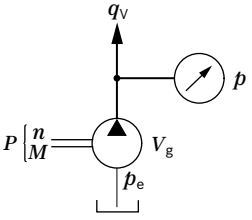


p_1 Maximum continuous pressure
p_2 Maximum intermittent pressure
p_3 Maximum pressure peak

Design calculations for pumps		
Flow	$q_v = \frac{V_g \times n \times \eta_v}{1000}$	[l/min]
Torque	$M = \frac{V_g \times \Delta p}{20 \times \pi \times \eta_{hm}}$	[Nm]
Power	$P = \frac{2 \pi \times M \times n}{60000} = \frac{q_v \times \Delta p}{600 \times \eta_t}$	[kW]
Pressure	$\Delta p = \frac{M \times 20 \times \pi \times \eta_{hm}}{V_g}$	[bar]
	$\Delta p = \frac{P \times 600 \times \eta_t}{q_v}$	[bar]
Displacement	$V_g = \frac{q_v \times 1000}{n \times \eta_v}$	[cm³]
	$V_g = \frac{M \times 20 \times \pi \times \eta_{hm}}{\Delta p}$	[cm³]
Rotational speed	$n = \frac{q_v \times 1000}{V_g \times \eta_v}$	[rpm]

Key

V_g	Displacement per revolution [cm³]
Δp	Differential pressure [bar] ($\Delta p = p - p_e$)
n	Rotational speed [rpm]
q_v	Flow [l/min]
M	Torque [Nm]
P	Power [kW]
η_v	Volumetric efficiency[1]
η_{hm}	Hydraulic-mechanical efficiency[1]
η_t	Total efficiency ($\eta_t = \eta_v \times \eta_{hm}$)[1]

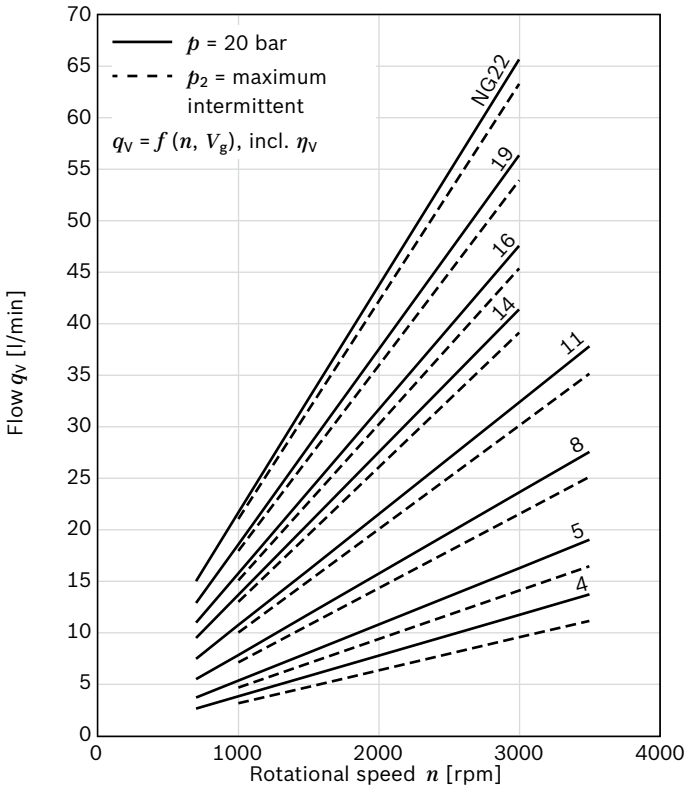


Note
On the following pages you can find diagrams for a rough calculation.

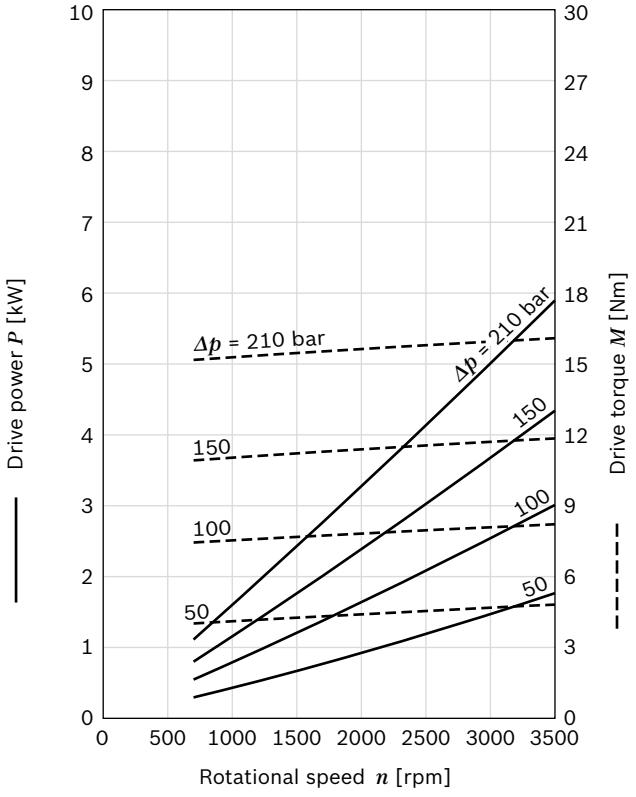
1) Parameter as a decimal, e.g. 0.9

Flow and power characteristic curves

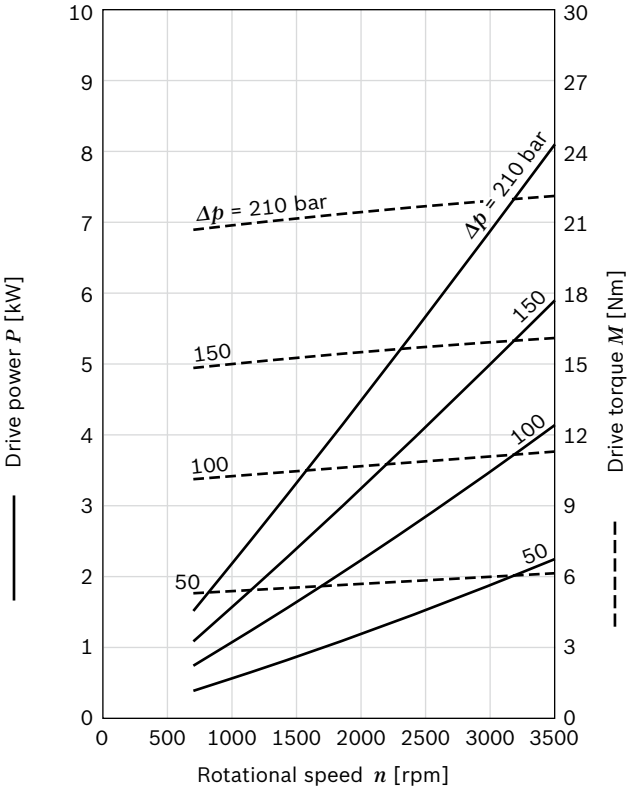
▼ **Flow**



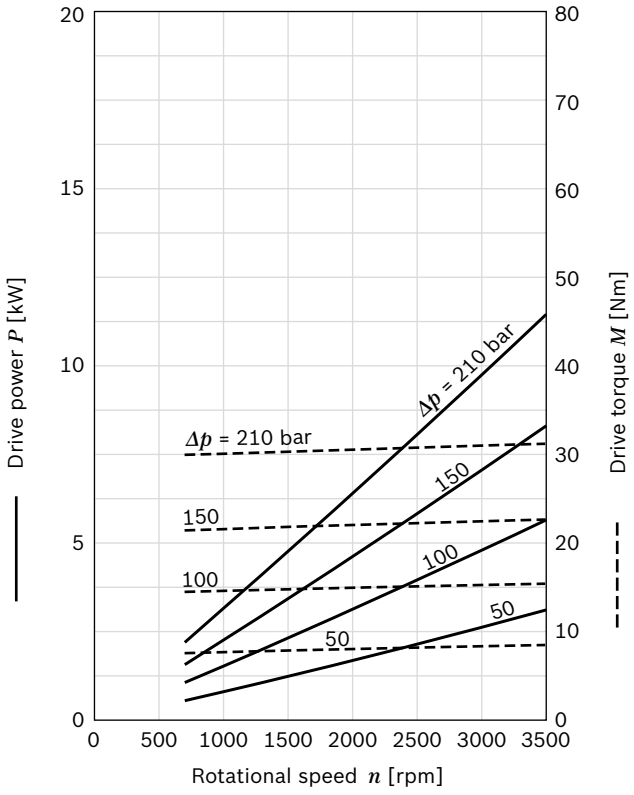
▼ **Size 4**



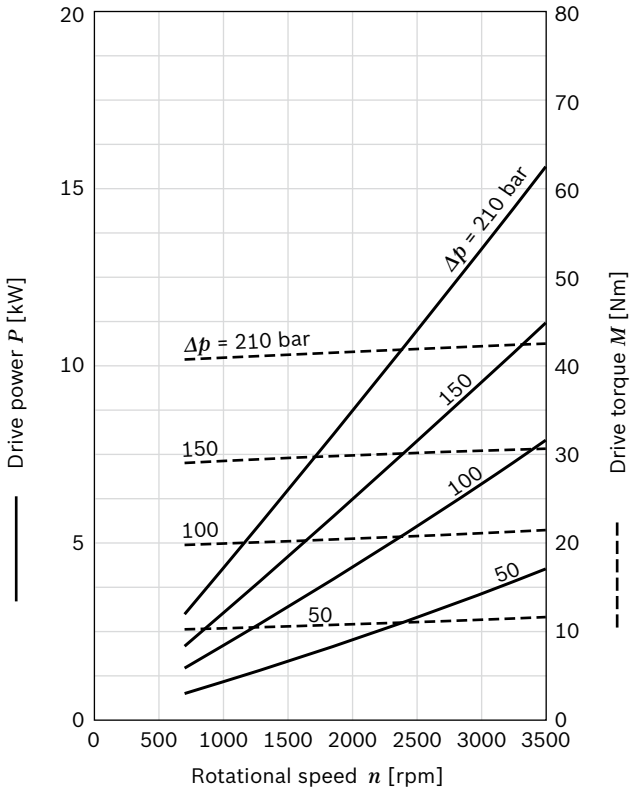
▼ **Size 5**



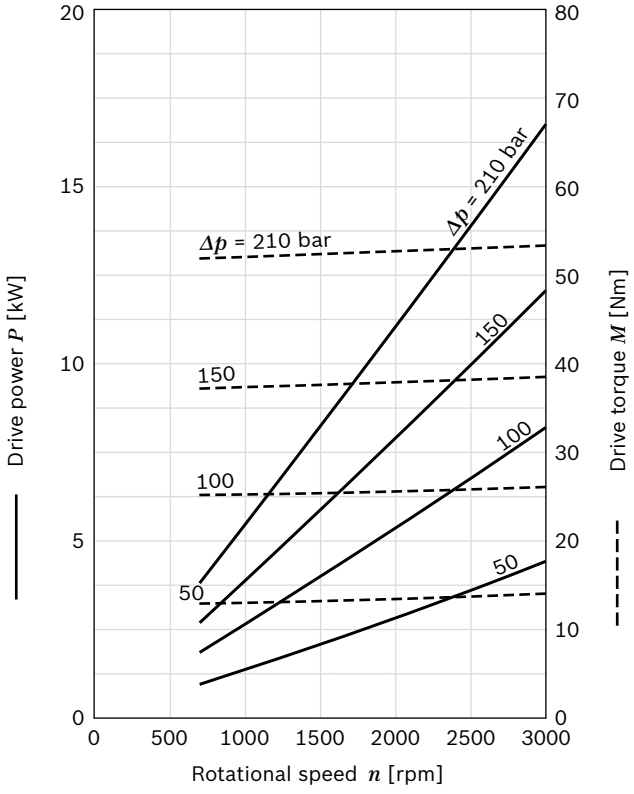
▼ **Size 8**



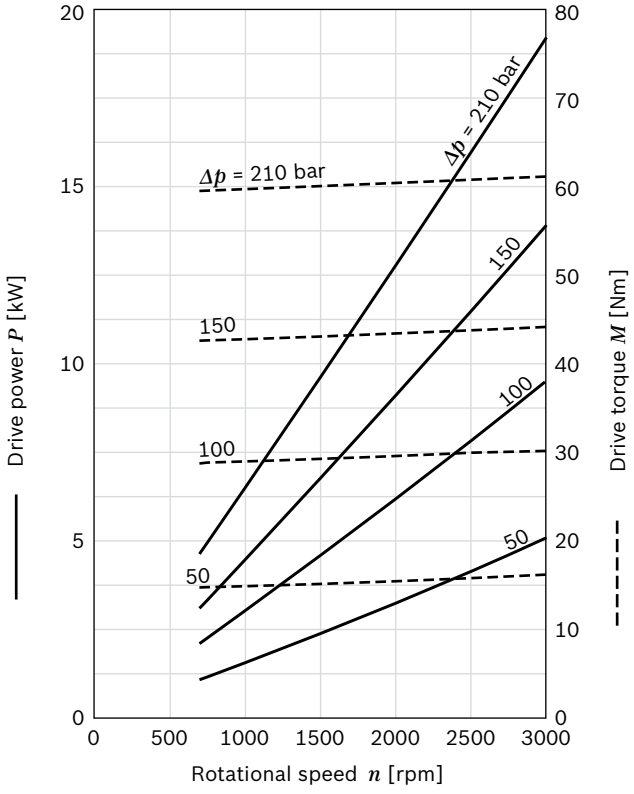
▼ **Size 11**



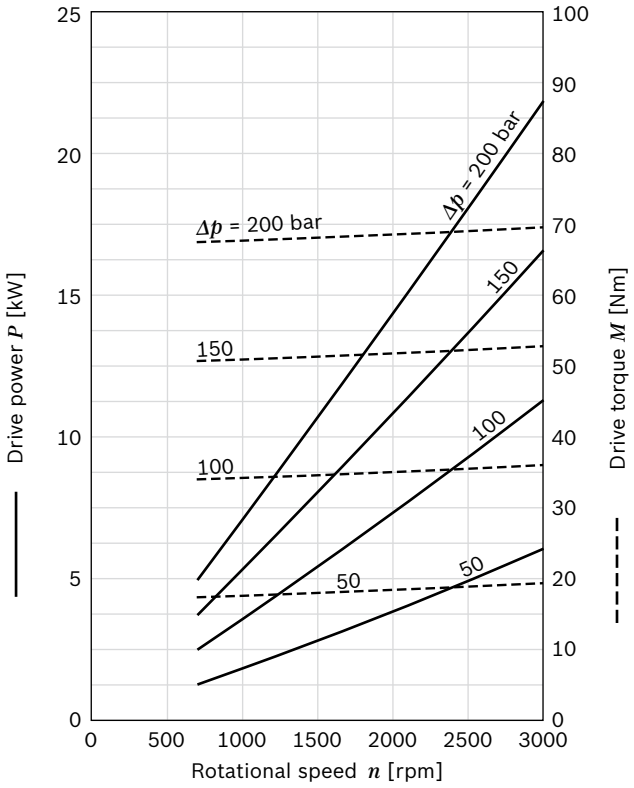
▼ **Size 14**



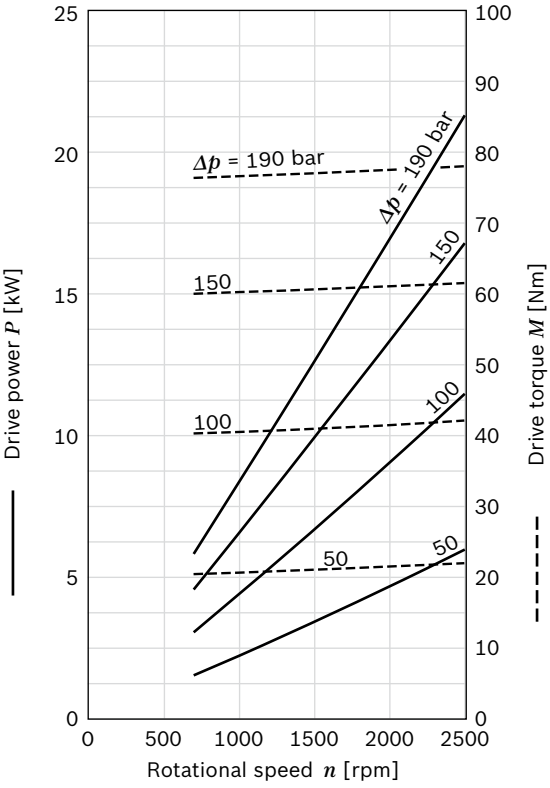
▼ **Size 16**



▼ **Size 19**



▼ **Size 22**



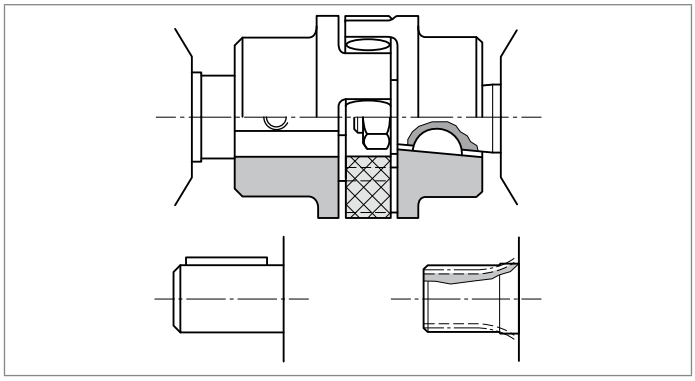
Note
Characteristic curves measured at ν = 32 mm²/s and θ = 50 °C.

$P = f(n, p)$, incl. η_t ——————
$M = f(n, p)$, incl. η_{hm} - - - - - -

Drives

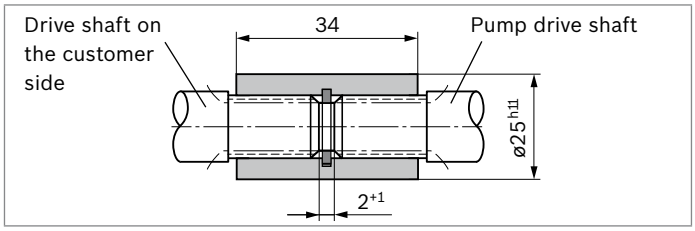
1. Elastic coupling

- The coupling must not transfer any radial or axial forces to the pump.
- The maximum radial runout from the shaft to the spigot may not exceed 0.3 mm.
- See the coupling manufacturer's assembly instructions for acceptable shaft misalignments.



2. Coupling sleeve

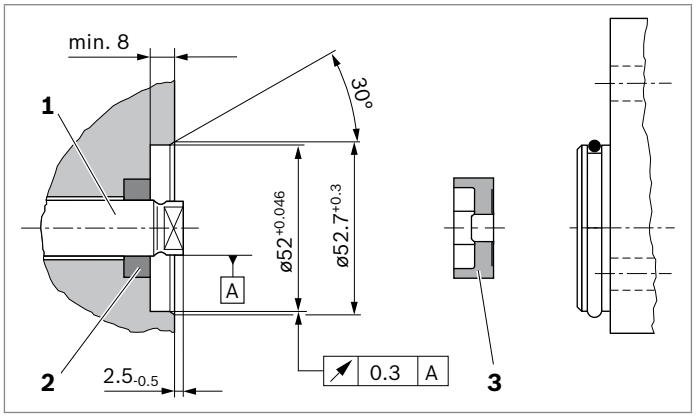
- To be used on splined shaft profile according to DIN and SAE
- Attention: No radial or axial forces are permitted on the pump shaft or coupling sleeve. The coupling sleeve must be free to move axially.
- The distance between the pump drive shaft and drive shaft on the customer side must be 2^{+1} mm
- Oil-bath or oil-mist lubrication is required



Drive shaft	M_{max} [Nm]	Size	p_{max} [bar]
F	100	4 to 16	210
		19	200
		22	190
R	110	4 to 16	210
		19	200
		22	190

3. Coupling dog

- For attaching the pump directly to an electric motor or combustion engine, gear, etc.
- The pump shaft has a special coupling dog and driver (**3**)
- No shaft seal
- Drive-side installation and sealing according to the following recommendations and dimensions



Drive shaft	M_{max} [Nm]	Size	p_{max} [bar]
N	52	4 to 14	210
		16	180
		19	150
		22	130

- **Drive shaft on the customer side (1)**
  - Case-hardening steel DIN 17210 e.g., 20 MnCrS 5 case-hardened 0.6 deep; HRC $60^{\pm 3}$
  - Seal ring running surface ground without rifling $R_{max} \leq 4\ \mu m$

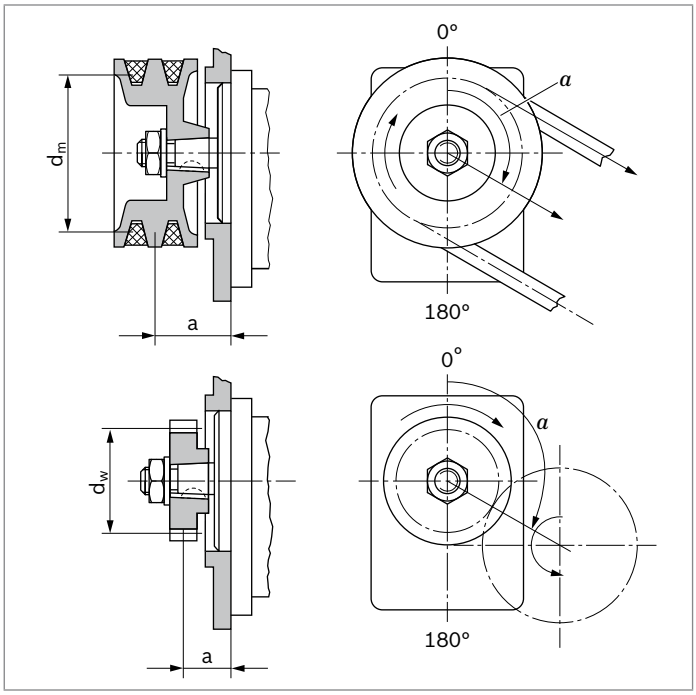
- **Radial shaft seal on the customer side (2)**
  - Provide with rubber cover (see DIN 3760, type AS, or double-lipped ring)
  - Provide installation edge with 15° slant or install shaft seal with protection sleeve

4. Tapered and keyed shafts

Drive shaft	M_{max} [Nm][1)]	Size	p_{max} [bar]
C	130	4 to 16	210
		19	200
		22	190
H	130	4 to 16	210
		19	200
		22	190
Q	55	4 to 14	210
		16	190
		19	165
		22	140

5. V-belts and straight gear wheels or helical toothed gear drives without outboard bearing

For V-belt or gear wheel drives, please contact us specifying the application and mounting conditions (dimensions a, d_m, d_w and angle α). For helical toothed gear drives, details of the helix angle β are also required.

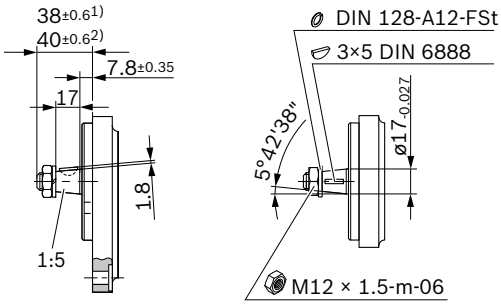


1) These values only apply while adhering to the defined conditions on page 6. Please consult Bosch Rexroth if the specified values are to be exceeded.

Dimensions – drive shaft

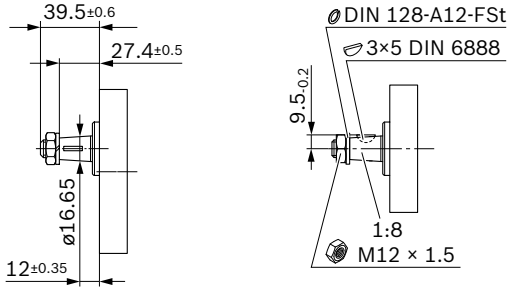
▼ **Tapered shaft 1:5**

C



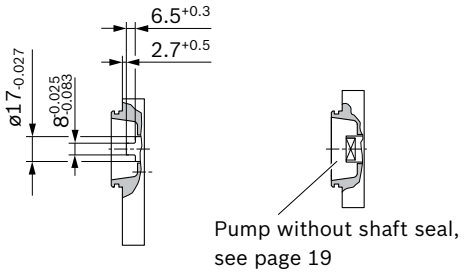
▼ **Tapered shaft 1:8**

H



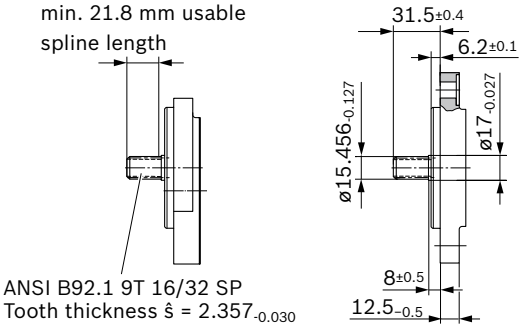
▼ **Dihedral claw**

N



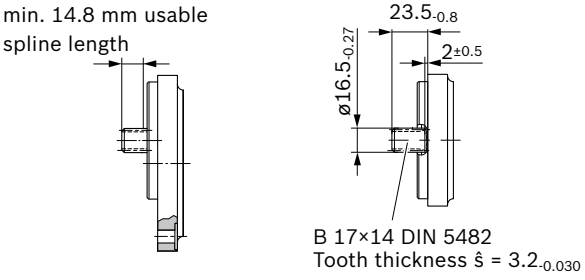
▼ **Splined shaft (SAE J744 16-4 9T)[3)]**

R



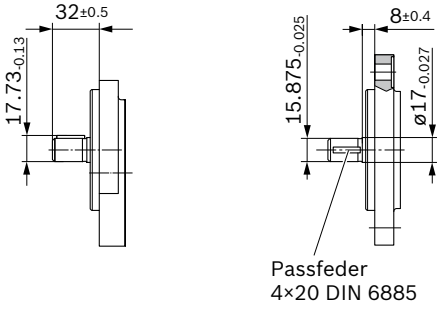
▼ **Splined shaft (DIN 5482 B17 × 14)**

F



▼ **Parallel keyed shaft (SAE J744 16-1 A)**

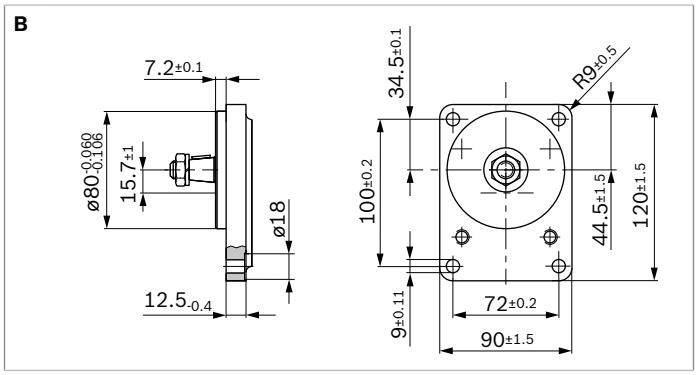
Q



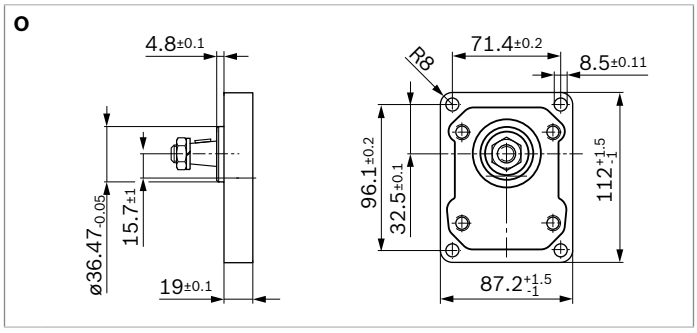
1) In combination with front cover B
2) In combination with front cover P and front cover N
3) For combination with an axial piston unit, see page 24 to 28

Dimensions – front cover

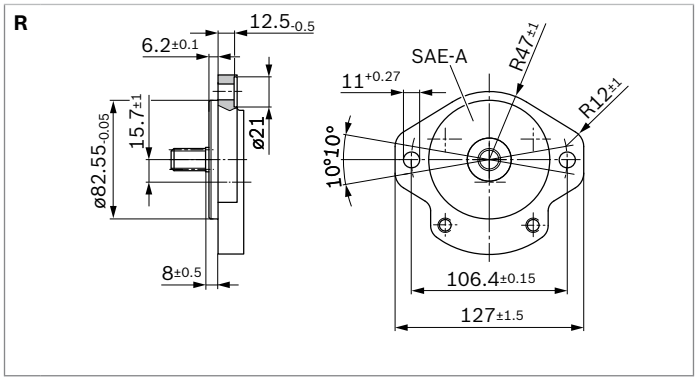
▼ **Rectangular flange Ø80 mm**



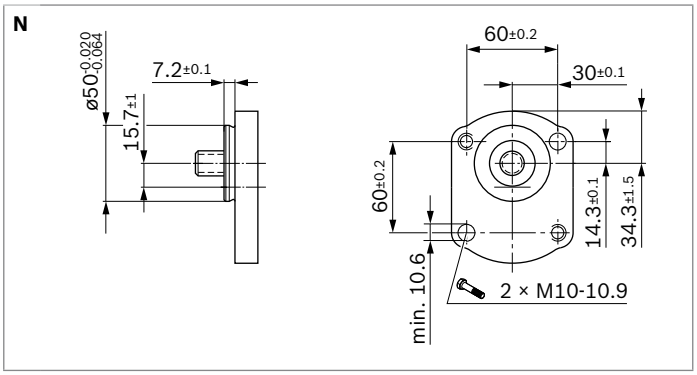
▼ **Rectangular flange Ø36.47 mm**



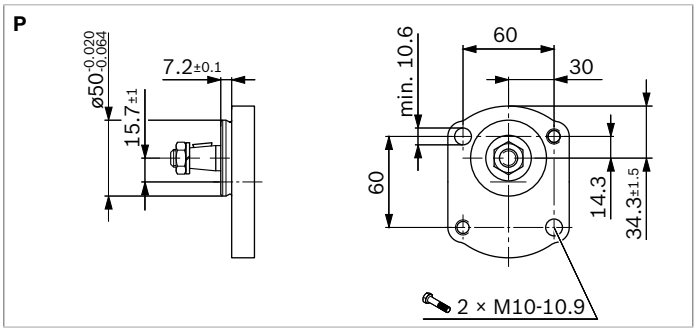
▼ **2-bolt flange Ø82.55 mm, SAE J744 82-2 A[1)]**



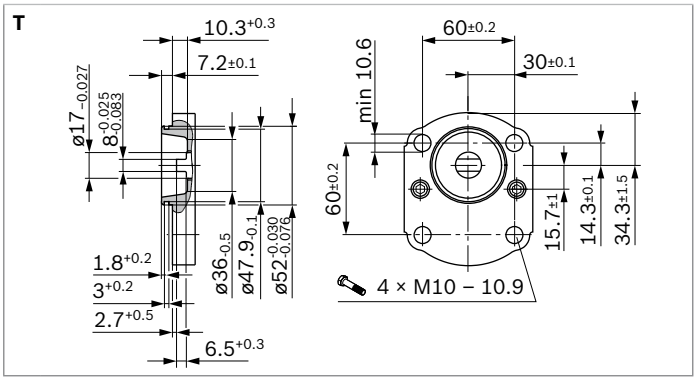
▼ **2-bolt mounting Ø50 mm, connection variant 1**



▼ **2-bolt mounting Ø50 mm, connection variant 2**



▼ **4-bolt mounting Ø = 52 mm with O-ring**

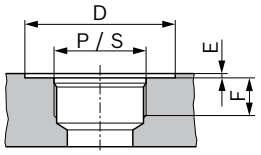


1) For combination with an axial piston unit, see page 24 to 28

Dimensions – line connection

▼ **Thread UNF-2B, SAE with O-ring**

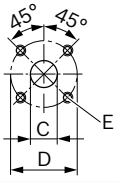
12



Size	Pressure side				Suction side			
	P	D	E	F	S	D	E	F
4 to 5	3/4-16 UNF-2B	30.2	0.5	14	7/8-14 UNF-2B	35	0.5	17
8 to 22	7/8-14 UNF-2B	35	0.5	17	1 1/16-12 UN-2B	45	0.5	19

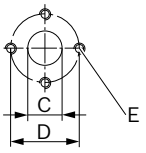
▼ **Square flange**

20



Size	Pressure side			Suction side		
	C	D	E	C	D	E
4 to 5	15	35	M6; 13 deep	15	40	M6; 13^{+1} deep
8 to 22	15	35	M6; 13 deep	20	40	M6; 13^{+1} deep

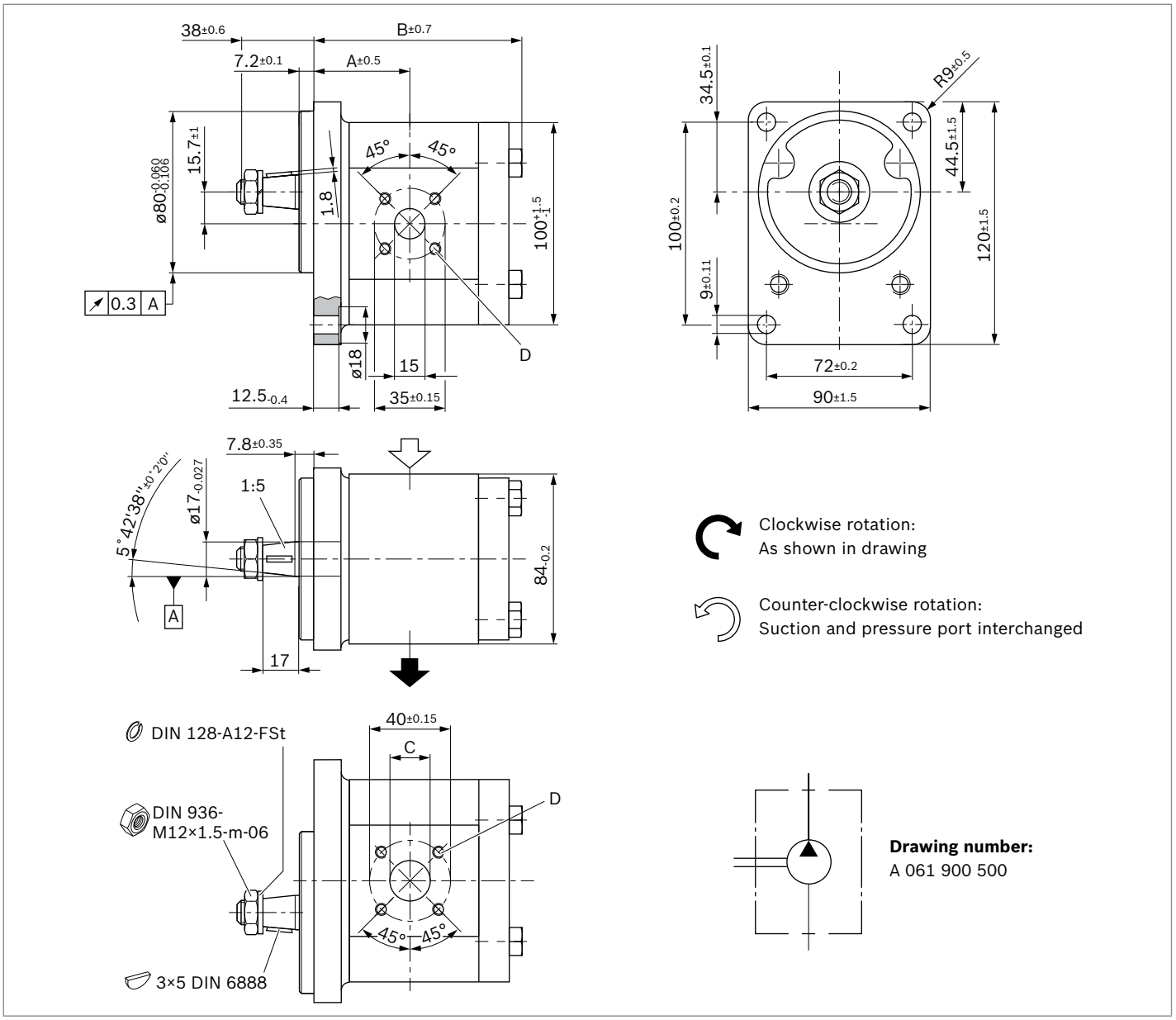
30



Size	Pressure side			Suction side		
	C	D	E	C	D	E
4 to 8	13.5	30.2	M6; 13 deep	13.5	30.2	M6; 13^{+1} deep
11 to 22	13.5	30.2	M6; 13 deep	20	39.7	M8; 13^{+1} deep

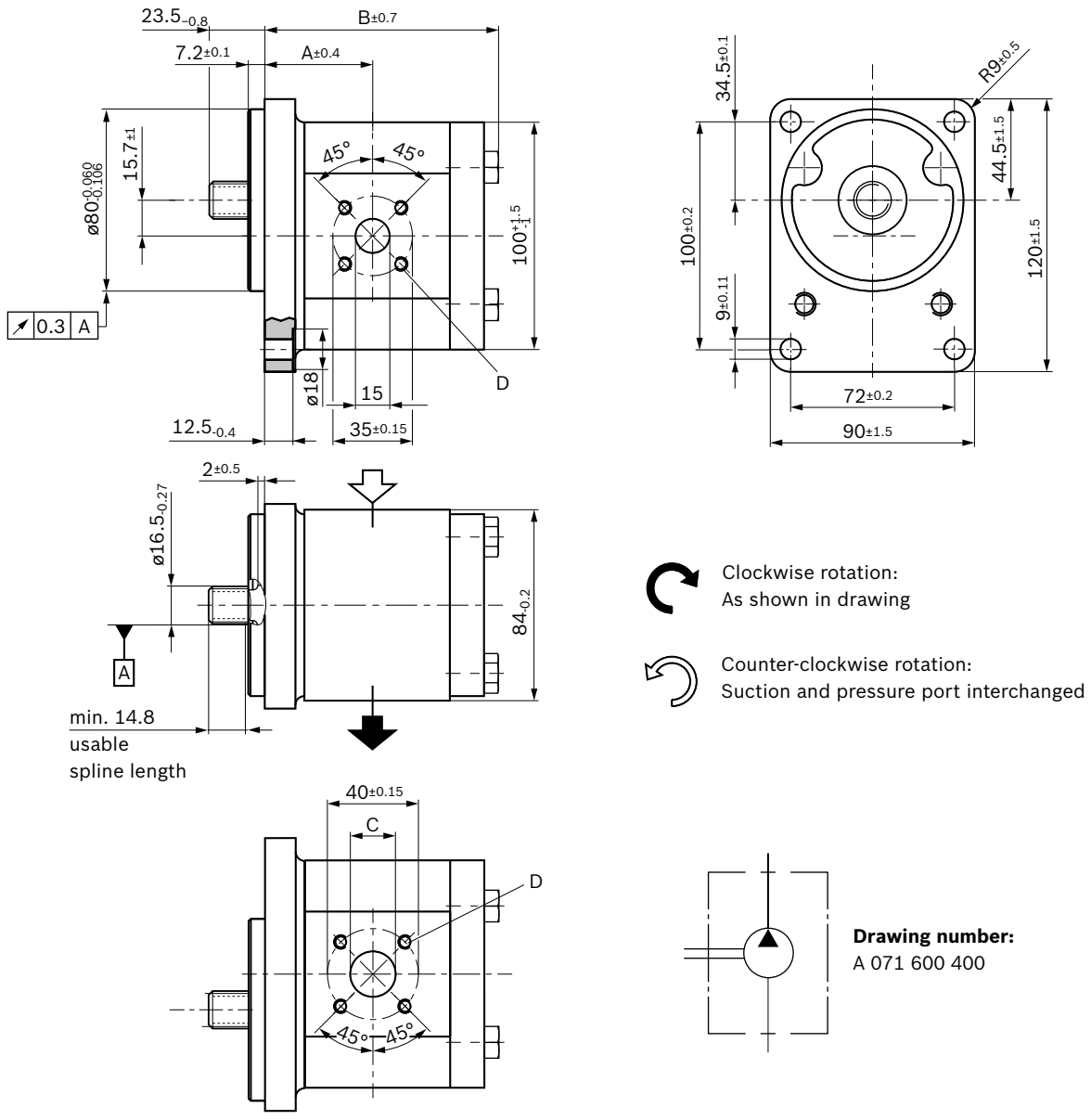
Dimensions – preferred series

▼ **Tapered shaft 1:5 with rectangular flange Ø80 mm**
AZPW–11– ... **CB20**MB



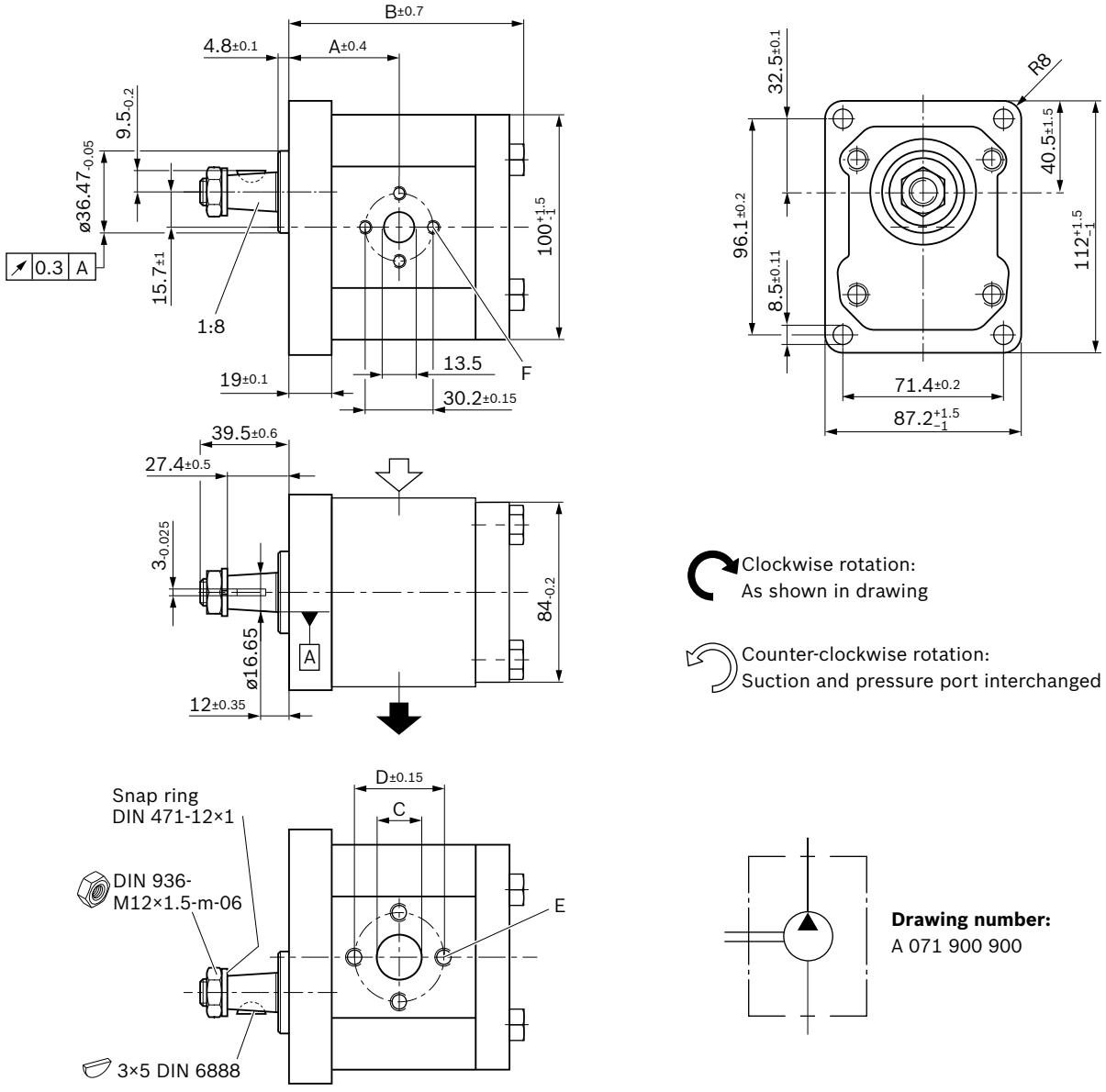
NG	Order number Direction of rotation Counter-clockwise	 Counter-clockwise	Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions A	 B	 C	 D
4	R983074473	R983074465	210	3500	39.9	83.8	15	M6; 13^{+1} deep
5	R983074474	R983074466	210	3500	41.1	86.3	15	
8	R983074475	R983074467	210	3500	43.2	90.4	20	
11	R983074476	R983074468	210	3500	47	95.4	20	
14	R983074477	R983074469	210	3000	47.5	100.4	20	
16	R983074478	R983074470	210	3000	47.5	103.8	20	
19	R983074479	R983074471	200	3000	47.5	108.8	20	
22	R983074480	R983074472	190	2500	55.1	114.2	20	

▼ **Splined shaft (DIN 5482 B17 × 14) with rectangular flange Ø80 mm**
AZPW–11 – … **FB20**MB



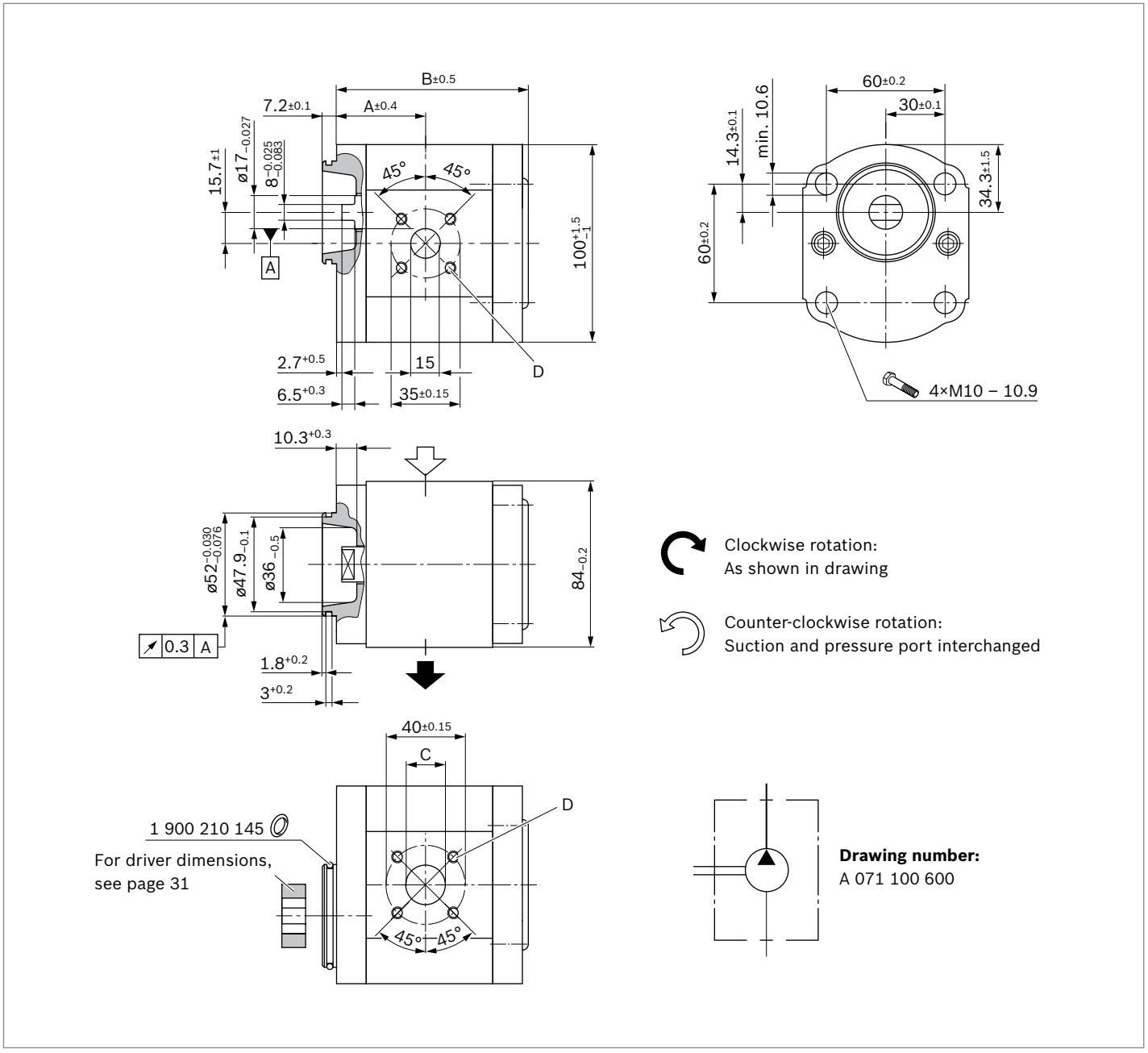
NG	Order number Direction of rotation Counter-clockwise	Clockwise	Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions A	B	C	D
4	R983072848	R983072840	210	3500	39.9	83.8	15	M6; 13^{+1} deep
5	R983072849	R983072841	210	3500	41.1	86.3	15	
8	R983072850	R983072842	210	3500	43.2	90.4	20	
11	R983072851	R983072843	210	3500	47	95.4	20	
14	R983072852	R983072844	210	3000	47.5	100.4	20	
16	R983072853	R983072845	210	3000	47.5	103.8	20	
19	R983072854	R983072846	200	3000	47.5	108.8	20	
22	R983072855	R983072847	190	2500	55.1	114.2	20	

▼ **Tapered shaft 1:8 with rectangular flange Ø36.47 mm**
AZPW–11 – ... **HO30**MB



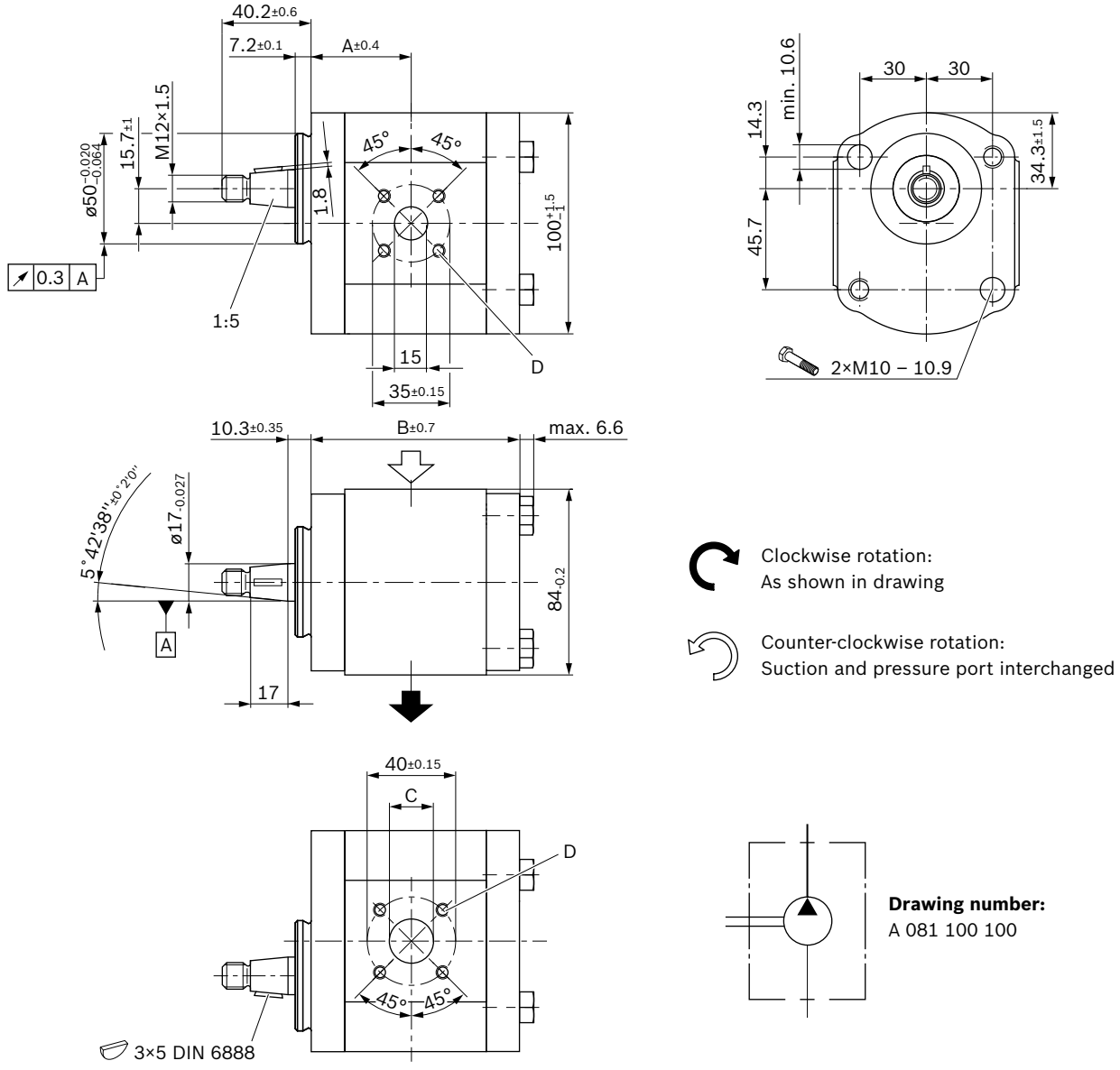
NG	Order number Direction of rotation		Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions					
	Counter-clockwise	Counter-clockwise			A	B	C	D	E	F
4	R983072831	R983072823	3500	210	41.4	85.4	13.5	30.2	M6; 13^{+1} deep	M6; 13^{+1} deep
5	R983072832	R983072824	3500	210	42.6	87.9	13.5	30.2		
8	R983072833	R983072825	3500	210	44.7	92	13.5	30.2		
11	R983072834	R983072826	3500	210	48.5	97	20	39.7	M8; 13^{+1} deep	
14	R983072835	R983072827	3000	210	49	102	20	39.7		
16	R983072836	R983072828	3000	210	49	105.4	20	39.7		
19	R983072837	R983072829	3000	200	49	110.4	20	39.7		
22	R983072838	R983072830	2500	190	56.6	115.8	20	39.7		

▼ **Dihedral claw with 4-bolt mounting Ø52 mm**
AZPW–11 – … NT20MB



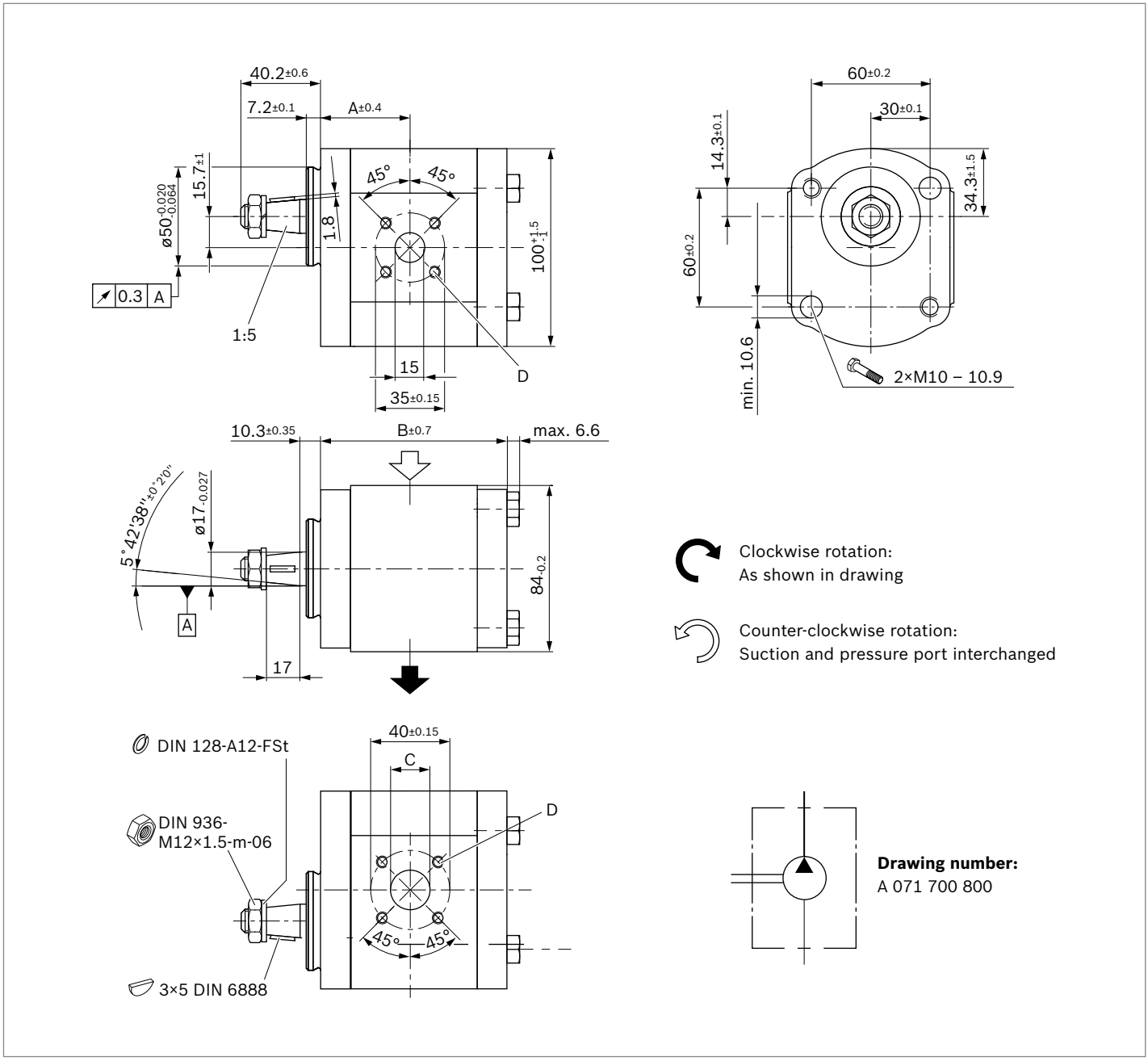
NG	Order number Direction of rotation		Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions			
	Counter-clockwise	Counter-clockwise			A	B	C	D
4	R983072863	R983072856	210	3500	37.4	75	15	M6; 13^{+1} deep
5	R983072864	R983076112	210	3500	38.6	77.5	15	
8	R983072865	R983072857	210	3500	40.7	81.6	20	
11	R983072866	R983072858	210	3500	44.5	86.6	20	
14	R983072867	R983072859	210	3000	45	91.6	20	
16	R983072868	R983072860	180	3000	45	95	20	
19	R983072869	R983072861	150	3000	45	100	20	
22	R983072870	R983072862	130	2500	52.6	105.4	20	

▼ **Tapered shaft 1:5 with 2-bolt flange Ø50 mm without nut and washer**
AZPW–11 – … **CP20**MB-**S0007**



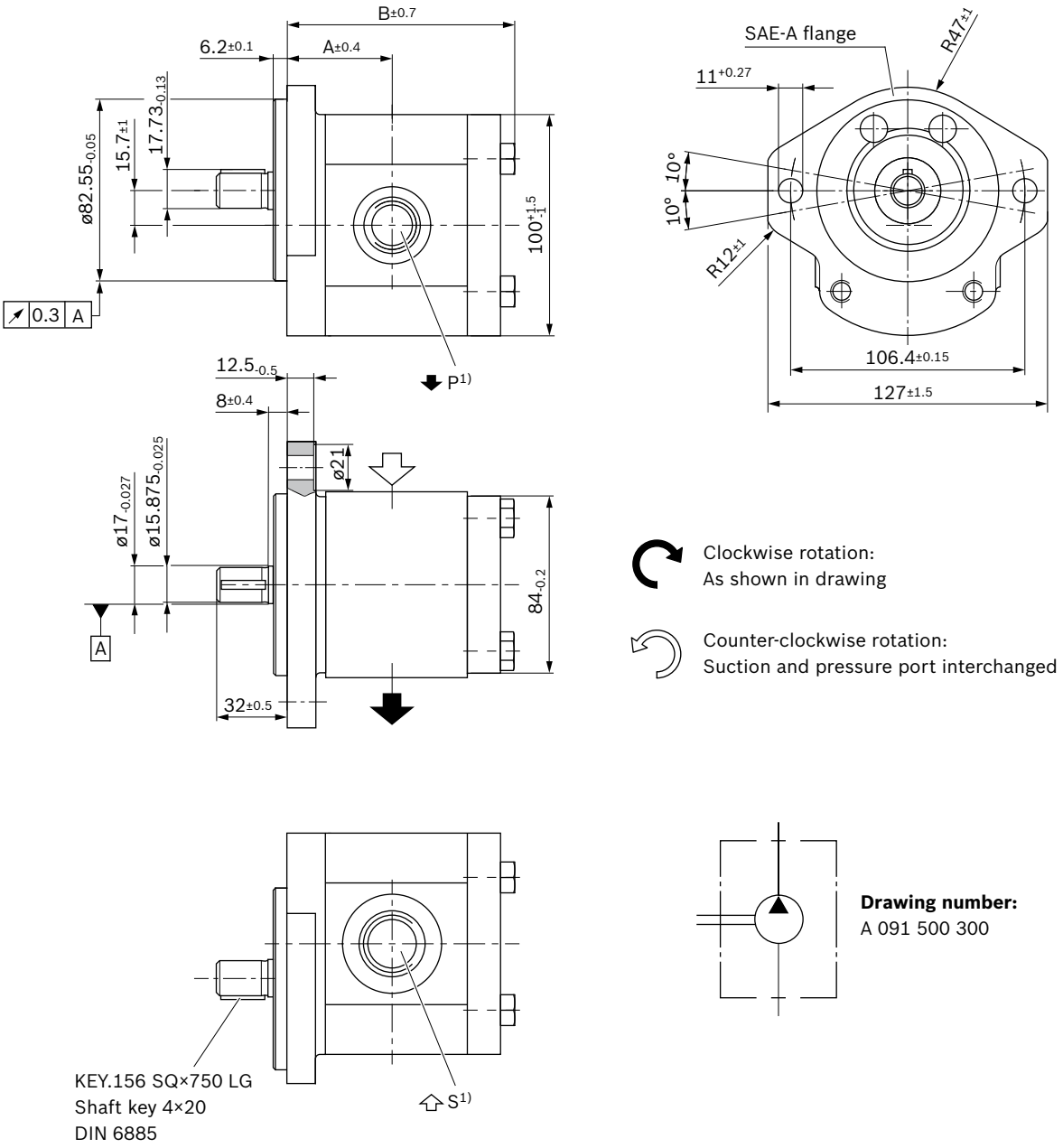
NG	Order number Direction of rotation Counter-clockwise	Clockwise	Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions A	B	C	D
4	R983072788	R983072780	210	3500	37.3	75	15	M6; 13^{+1} deep
5	R983072789	R983072781	210	3500	38.6	77.5	15	
8	R983072790	R983072782	210	3500	40.6	81.6	20	
11	R983072791	R983072783	210	3500	44.5	86.6	20	
14	R983072792	R983072784	210	3000	45	91.6	20	
16	R983072793	R983072785	210	3000	45	95	20	
19	R983072794	R983072786	200	3000	45	100	20	
22	R983072795	R983072787	190	2500	52.6	105.4	20	

▼ **Tapered shaft 1:5 with 2-bolt flange Ø50 mm**
AZPW–11 – … **CN20**MB



NG	Order number Direction of rotation Counter-clockwise	Clockwise	Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions A	B	C	D
4	R983072886	R983072878	210	3500	37.3	75	15	M6; 13^{+1} deep
5	R983072887	R983072879	210	3500	38.6	77.5	15	
8	R983072888	R983072880	210	3500	40.6	81.6	20	
11	R983072889	R983072881	210	3500	44.5	86.6	20	
14	R983072890	R983072882	210	3000	45	91.6	20	
16	R983072891	R983072883	210	3000	45	95	20	
19	R983072892	R983072884	200	3000	45	100	20	
22	R983072893	R983072885	190	2500	52.6	105.4	20	

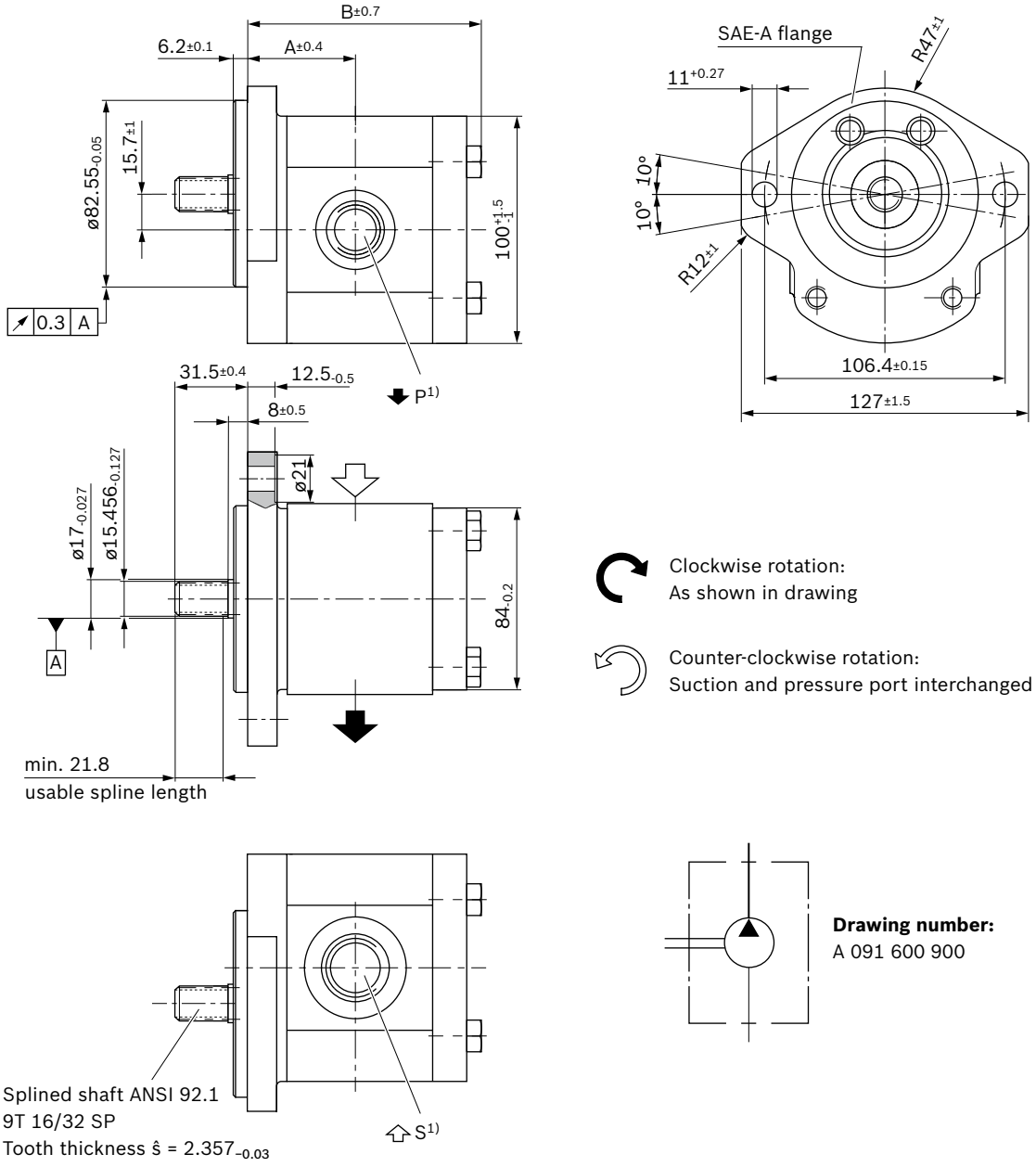
▼ **Parallel keyed shaft (SAE J744 16-1 A) with 2-bolt flange Ø82.55 mm (SAE J744 82-2 A)**
AZPW–11 – ... **QR12**MB



1) For dimension please see page 15

NG	Order number Direction of rotation		Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions			
	Counter-clockwise	Clockwise			A	B	P	S
4	R983072763	R983072755	210	3500	39.9	83.8	3/4-16 UNF-2B	7/8-14 UNF-2B
5	R983072765	R983072756	210	3500	41.1	86.3	3/4-16 UNF-2B	7/8-14 UNF-2B
8	R983072766	R983072757	210	3500	43.2	90.4	7/8-14 UNF-2B	1 1/16-12 UN-2B
11	R983072767	R983072758	210	3500	47	95.4	7/8-14 UNF-2B	1 1/16-12 UN-2B
14	R983072768	R983072759	210	3000	47.5	100.4	7/8-14 UNF-2B	1 1/16-12 UN-2B
16	R983072769	R983072760	190	3000	47.5	103.8	7/8-14 UNF-2B	1 1/16-12 UN-2B
19	R983072770	R983072761	165	3000	47.5	108.8	7/8-14 UNF-2B	1 1/16-12 UN-2B
22	R983072772	R983072762	140	2500	55.1	114.2	7/8-14 UNF-2B	1 1/16-12 UN-2B

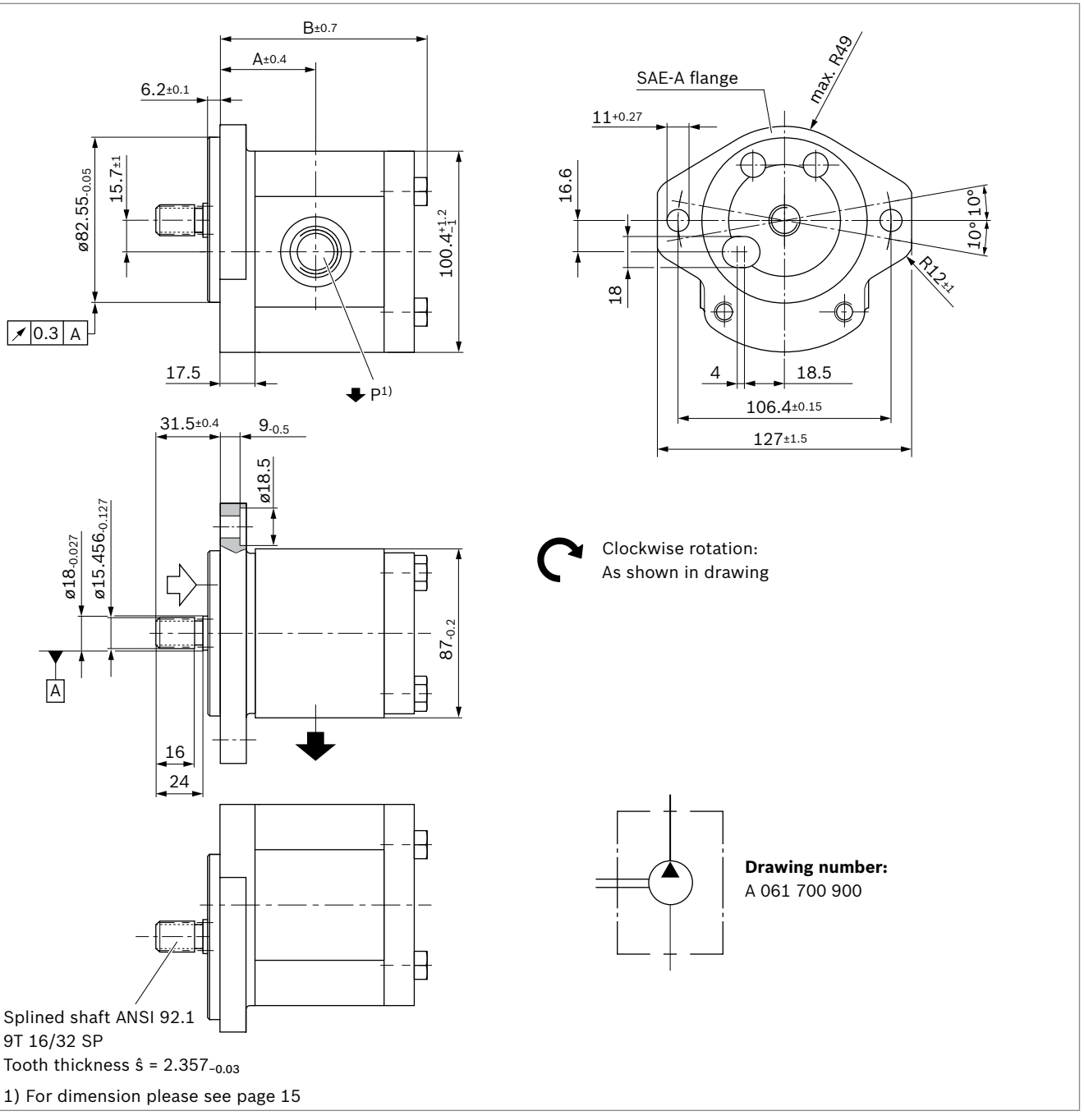
▼ **Splined shaft (SAE J744 16-4 9T) with 2-bolt flange Ø82.55 mm (SAE J744 82-2 A)**
AZPW–11 – ... **RR12**MB



1) For dimension please see page 15

NG	Order number Direction of rotation		Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions			
	Counter-clockwise	Clockwise			A	B	P	S
4	R983075416	R983075404	210	3500	39.9	83.8	3/4-16 UNF-2B	7/8-14 UNF-2B
5	R983075417	R983075405	210	3500	41.1	86.3	3/4-16 UNF-2B	7/8-14 UNF-2B
8	R983075418	R983075406	210	3500	43.2	90.4	7/8-14 UNF-2B	1 1/16-12 UN-2B
11	R983075419	R983075409	210	3500	47	95.4	7/8-14 UNF-2B	1 1/16-12 UN-2B
14	R983075420	R983075412	210	3000	47.5	100.4	7/8-14 UNF-2B	1 1/16-12 UN-2B
16	R983075421	R983075413	210	3000	47.5	103.8	7/8-14 UNF-2B	1 1/16-12 UN-2B
19	R983075422	R983075414	200	3000	47.5	108.8	7/8-14 UNF-2B	1 1/16-12 UN-2B
22	R983075423	R983075415	190	2500	55.1	114.2	7/8-14 UNF-2B	1 1/16-12 UN-2B

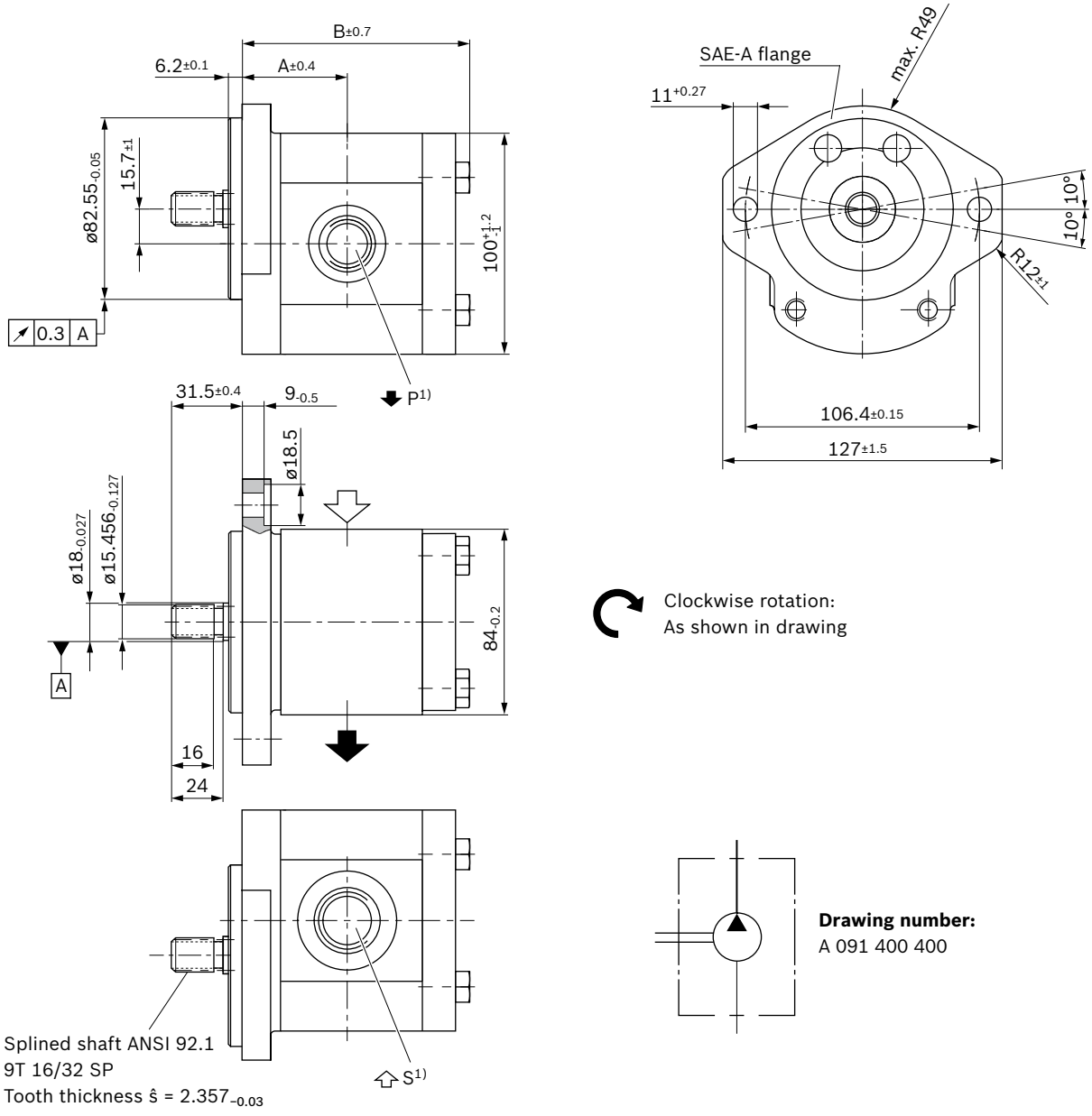
▼ **Splined shaft (SAE J744 16-4 9T) with 2-bolt flange Ø82.55 mm (SAE J744 82-2 A) for mounting on an axial piston unit and suction port in the front cover**
AZPW–11 – … **RR12**PB-**S0036**



1) For dimension please see page 15

NG	Order number Direction of rotation Clockwise	Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions A	B	P
5	R983074188	210	3500	41.2	86.3	3/4-16 UNF-2B
8	R983067892	210	3500	43.2	90.4	7/8-14 UNF-2B
11	R983074189	210	3500	45.6	95.4	7/8-14 UNF-2B
14	R983070537	210	3000	48.4	100.4	7/8-14 UNF-2B
16	R983074190	210	3000	49.9	103.8	7/8-14 UNF-2B
19	R983064536	200	2900	53.9	108.8	7/8-14 UNF-2B
22	R983074191	190	2400	55	114.2	7/8-14 UNF-2B

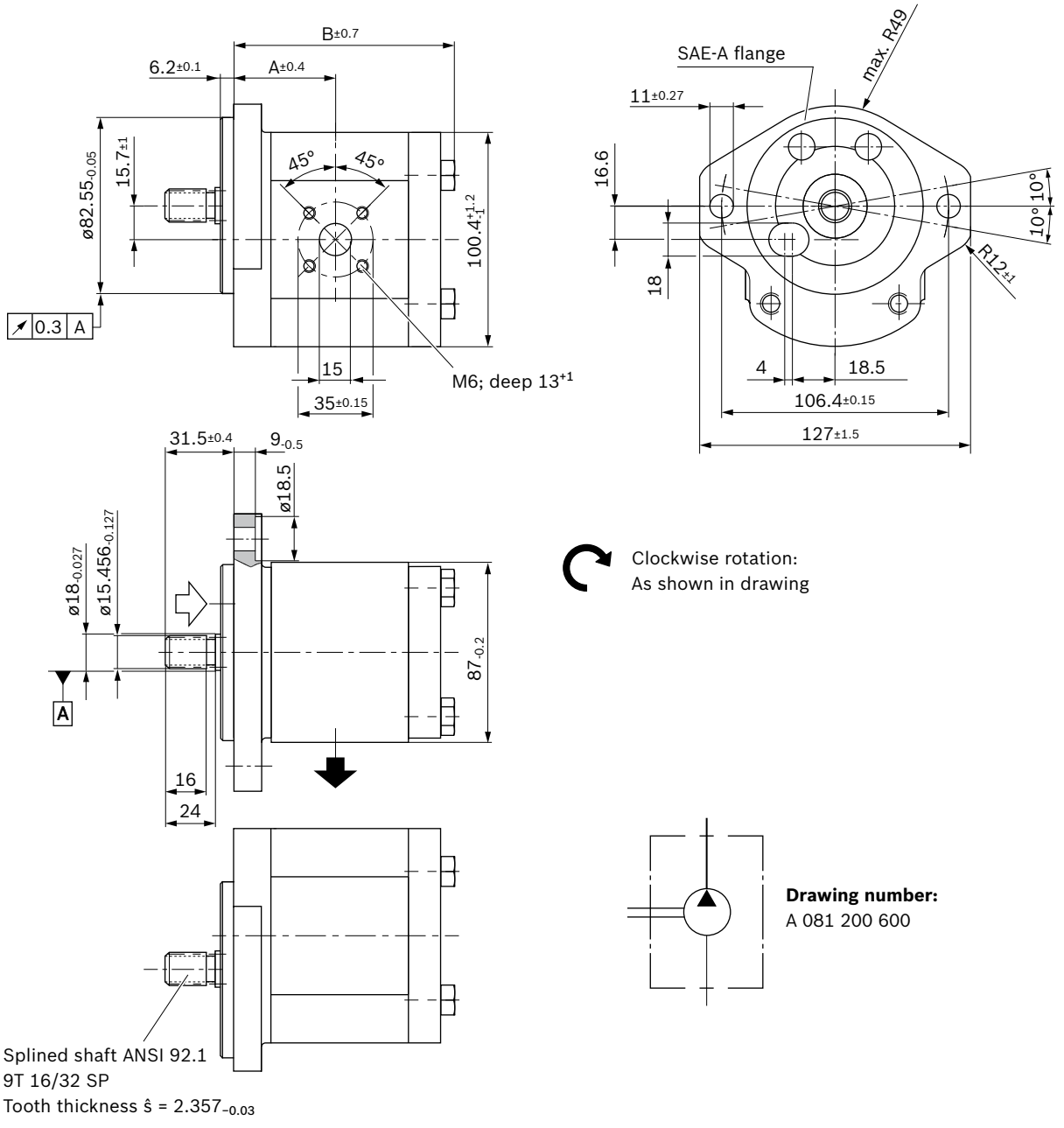
▼ **Splined shaft (SAE J744 16-4 9T) with 2-bolt flange Ø82.55 mm (SAE J744 82-2 A) for mounting on an axial piston unit**
AZPW–11 – ... **RR12**PB-**S0081**



1) For dimension please see page 15

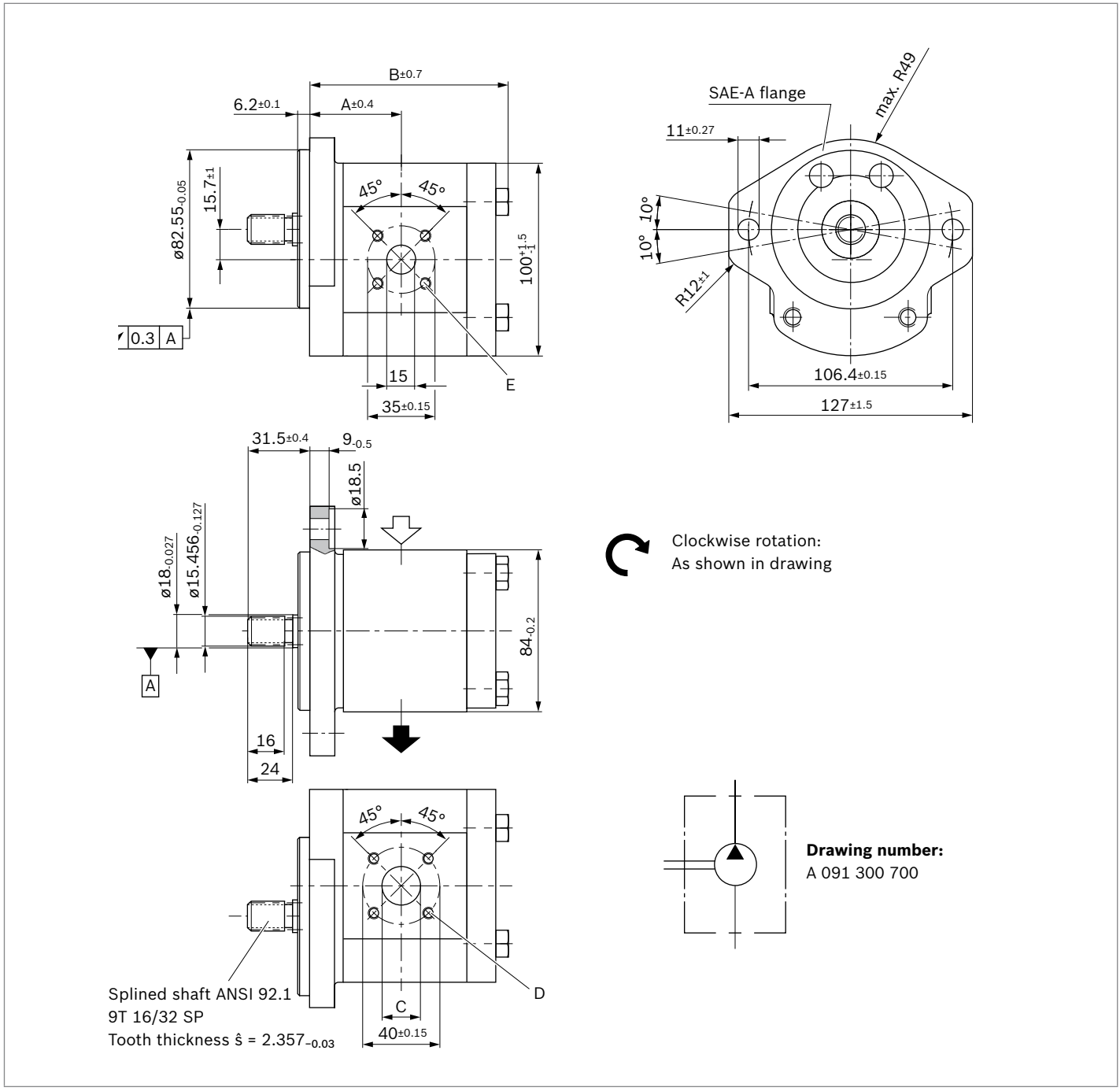
NG	Order number Direction of rotation Clockwise	Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions A	B	P	S
5	R983072871	210	3500	41.1	86.4	3/4-16 UNF-2B	7/8-14 UNF-2B
8	R983072872	210	3500	43.2	90.5	7/8-14 UNF-2B	1 1/16-12 UN-2B
11	R983072873	210	3500	45.7	95.5	7/8-14 UNF-2B	1 1/16-12 UN-2B
14	R983072874	210	3000	48.2	100.5	7/8-14 UNF-2B	1 1/16-12 UN-2B
16	R983072875	210	3000	49.9	103.9	7/8-14 UNF-2B	1 1/16-12 UN-2B
19	R983072876	200	3000	52.4	108.9	7/8-14 UNF-2B	1 1/16-12 UN-2B
22	R983072877	190	2500	55.1	114.3	7/8-14 UNF-2B	1 1/16-12 UN-2B

▼ **Splined shaft (SAE J744 16-4 9T) with 2-bolt flange Ø82.55 mm (SAE J744 82-2 A) for mounting on an axial piston unit and suction port in the front cover**
AZPW–11 – … **RR20**PB-**S0036**



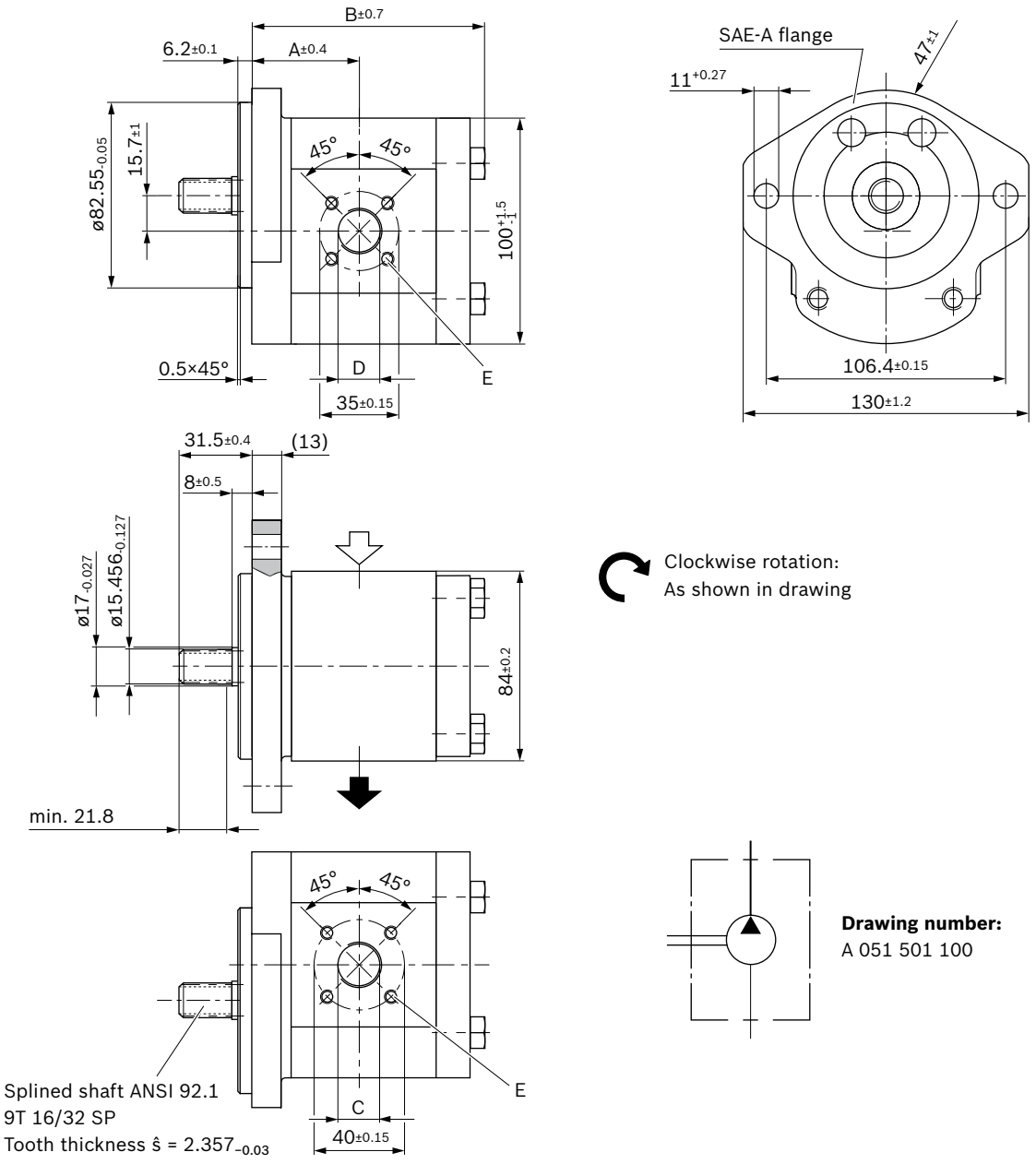
NG	Order number Direction of rotation Clockwise	Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions A	 B
5	R983072894	210	3500	41.1	86.3
8	R983072895	210	3500	43.2	90.4
11	R983072896	210	3500	47	95.4
14	R983072897	210	3500	47.5	100.4
16	R983072898	210	3500	47.5	103.8
19	R983072899	200	2900	47.5	108.8
22	R983072900	190	2400	55.1	114.2

▼ **Splined shaft (SAE J744 16-4 9T) with 2-bolt flange Ø82.55 mm (SAE J744 82-2 A) for mounting on an axial piston unit**
AZPW–11 – … **RR20**PB-**S0081**



NG	Order number Direction of rotation Clockwise	Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions A	B	C	D	E
5	R983077585	210	3500	41.1	86.3	15	M6; 13^{+1} deep	M6; 13^{+1} deep
8	R983077586	210	3500	43.2	90.4	20		
11	R983077587	210	3500	47	95.4	20		
14	R983077588	210	3500	47.5	100.4	20		
16	R983077589	210	3500	47.5	103.8	20		
19	R983077590	200	3300	47.5	108.8	20	M8; 13^{+1} deep	
22	R983077591	190	3200	55.1	114.2	20		

▼ **Splined shaft (SAE J744 16-4 9T) with 2-bolt flange Ø82.55 mm (SAE J744 82-2 A) for mounting on an axial piston unit and line connections in Non-standard version**
AZPW–21 – ... **RRXX**MB-**S0593**

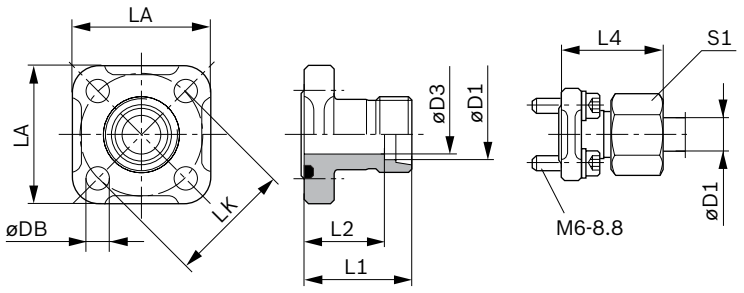


NG	Order number Direction of rotation Clockwise	Maximum intermittent pressure p_2 [bar]	Maximum rotational speed [rpm]	Dimensions				
				A	B	C	D	E
4	R983035496	210	3500	39.9	84.7	3/8; 15 deep	3/8; 15 deep	M6; min. 13 deep
5	R983035497	210	3500	41.1	87.3	3/8; 15 deep	3/8; 15 deep	
8	R983035498	210	3500	43.2	91.4	1/2; 15 deep	1/2; 15 deep	
11	R983035499	210	3500	47	95.7	1/2; 19 deep	1/2; 15 deep	
14	R983035500	210	3500	47.5	101.4	3/4; 19 deep	1/2; 15 deep	
16	R983035501	210	3500	47.5	104.8	3/4; 19 deep	1/2; 15 deep	
19	R983035502	200	3300	47.5	109.8	3/4; 19 deep	1/2; 15 deep	
22	R983035503	190	3200	55.1	115.2	3/4; 19 deep	1/2; 15 deep	

Accessories

Gear pump flanges, straight, for square flange 20 (see page 14)

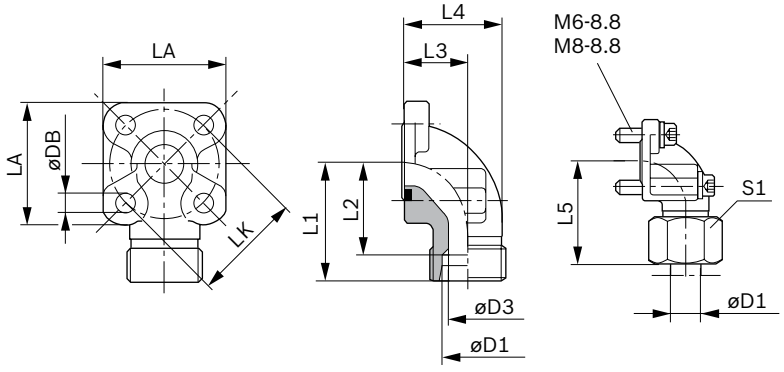
Complete fitting with O-ring, metric screw set, nuts and olive.



LK	D1	D3	L1	L2	L4	LA	1	DB	Screws 4 pcs.	O-ring NBR	Weight [kg]	Order number	*p* [bar]
35	10L	8	30	23.0	39.0	40	19	6.4	M6 × 22	20 × 2.5	0.09	1 515 702 064	315
35	12L	10	30	23.0	39.0	40	22	6.4	M6 × 22	20 × 2.5	0.10	1 515 702 065	315
35	15L	12	30	23.0	38.0	40	27	6.4	M6 × 22	20 × 2.5	0.10	1 515 702 066	250
40	15L	12	35	28.0	43.0	42	27	6.4	M6 × 22	24 × 2.5	0.12	1 515 702 067	100
40	18L	15	35	27.5	44.0	42	32	6.4	M6 × 22	24 × 2.5	0.13	1 515 702 068	100
40	22L	19	35	27.5	44.5	42	36	6.4	M6 × 22	24 × 2.5	0.12	1 515 702 069	100
40	28L	24	42	27.5	34.5	42	41	6.4	M6 × 22	24 × 2.5	0.15	1 515 702 008	100

Gear pump flanges, 90° angle, for square flange 20 (see page 14)

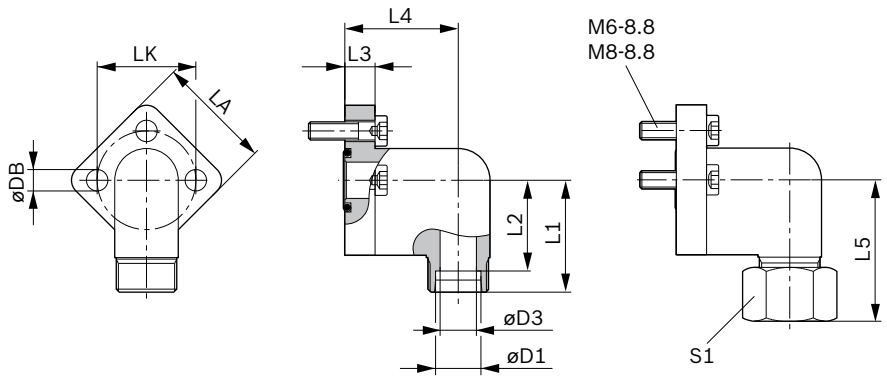
Complete fitting with O-ring, metric screw set, nuts and olive.



LK	D1	D3	L1	L2	L3	L4	L5	LA	1	DB	Screws 2 pcs.	2 pcs.	O-ring NBR	Weight [kg]	Order number	*p* [bar]
35	10L	8	38	31.0	16.5	26.5	47.0	40	19	6.4	M6 × 22	M6 × 35	20 × 2.5	0.16	1 515 702 070	315
35	12L	10	38	31.0	16.5	26.5	47.0	40	22	6.4	M6 × 22	M6 × 35	20 × 2.5	0.16	1 515 702 071	315
35	15L	12	38	31.0	16.5	26.5	46.0	40	27	6.4	M6 × 22	M6 × 35	20 × 2.5	0.15	1 515 702 072	250
35	16S	12	38	29.5	20.0	31.0	48.0	40	30	6.4	M6 × 22	M6 × 40	20 × 2.5	0.18	1 515 702 002	315
35	18L	15	38	29.5	20.0	31.0	47.0	40	32	6.4	M6 × 22	M6 × 40	20 × 2.5	0.18	1 545 702 006	250
35	20S	16	45	34.5	25.0	38.0	56.0	40	36	6.4	M6 × 22	M6 × 45	20 × 2.5	0.24	1 515 702 017	315
40	15L	12	38	31.0	22.5	36.5	46.0	42	27	6.4	M6 × 22	M6 × 22	24 × 2.5	0.15	1 515 702 073	100
40	18L	15	38	30.5	22.5	36.5	47.0	42	32	6.4	M6 × 22	M6 × 22	24 × 2.5	0.17	1 515 702 074	100
40	20S	16	40	29.5	22.5	35.5	50.0	42	36	6.4	M6 × 22	M6 × 45	24 × 2.5	0.20	1 515 702 011	250
40	22L	19	38	30.5	22.5	36.5	47.5	42	36	6.4	M6 × 22	M6 × 22	24 × 2.5	0.17	1 515 702 075	100
40	28L	22	40	32.5	28.0	43.0	49.0	42	41	6.4	M6 × 20	M6 × 50	24 × 2.5	0.24	1 515 702 010	100
40	35L	31	41	30.5	34.0	55.0	52.0	42	50	6.4	M6 × 22	M6 × 60	24 × 2.5	0.33	1 515 702 018	100

Gear pump flange, 3-hole, 90° angle, for square flange 30 (see page 14)

Complete fitting with
O-ring, metric screw set,
nuts and olive.

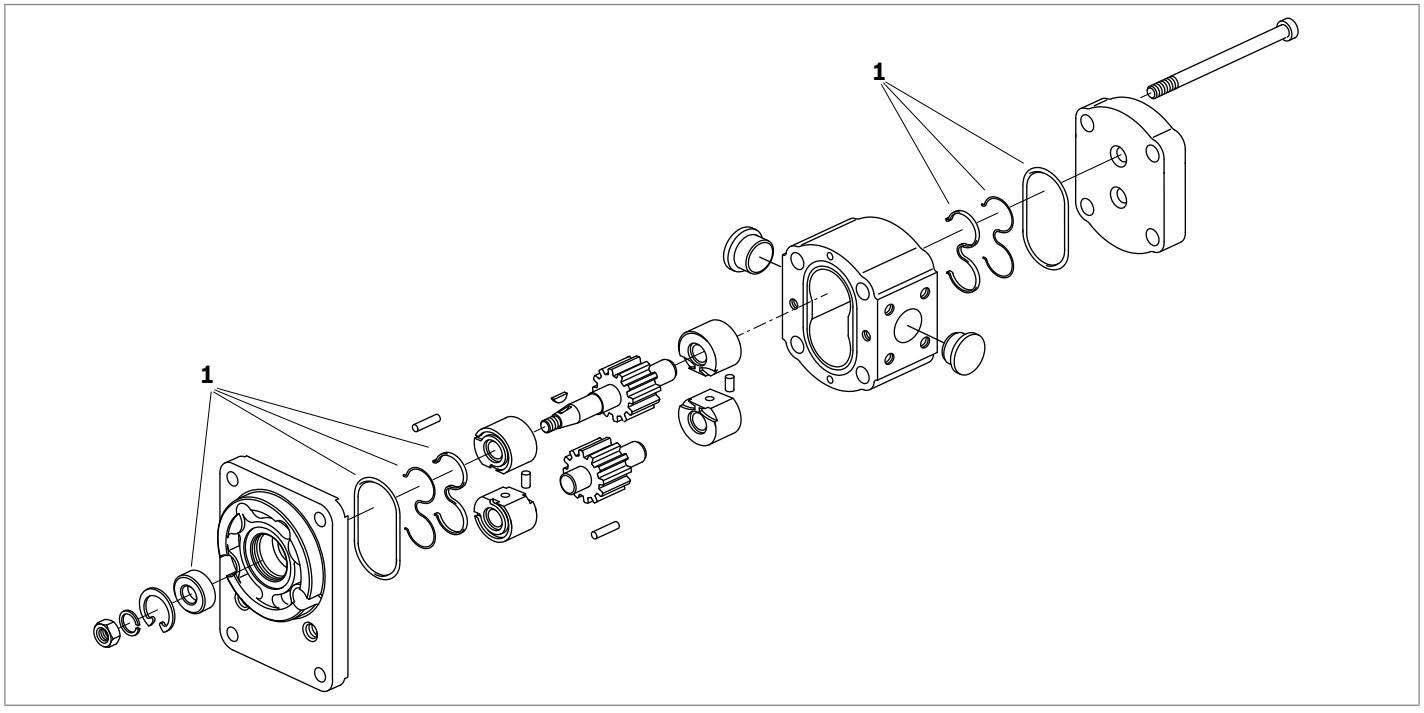


LK	D1	D3	L1	L2	L3	L4	L5	LA	1	DB	Screws 3 pcs.	O-ring NBR	Weight [kg]	Order number	p [bar]
30	12L	10	37	30.0	10	37.5	46	38	22	6.4	M6 × 22	16 × 2.5	0.13	1 515 702 146	250
30	15L	12	37	30.0	10	37.5	47	38	27	6.4	M6 × 22	16 × 2.5	0.14	1 515 702 147	250
30	18L	15	37	30.0	10	37.5	47	38	32	6.4	M6 × 22	16 × 2.5	0.17	1 515 702 148	160
40	22L	19	43	35.5	14	41.0	53	48	36	8.4	M8 × 30	24 × 2.5	0.29	1 515 702 149	160
40	28L	24	43	35.5	14	41.0	53	48	41	8.4	M8 × 30	24 × 2.5	0.40	1 515 702 150	160

Note
You can find the permissible tightening torques in our publication 07012-B1 “General Instruction Manual for External Gear Units”.

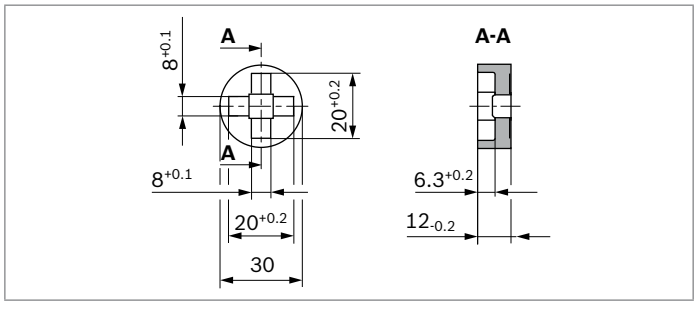
Spare parts

▼ **Schematic diagram**

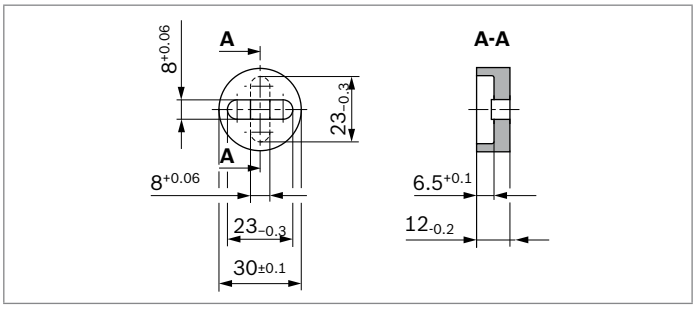


Item	Designation	Order number	Material	AZPW–22 – …								
				CB…	FB…	HO…	NT…	CP…	CN…	QR…	RR12	RR20
1	Seal kit	R 983 032 388	NBR	x	x	x	x	x	x	x	x	
		R 983 069 216	FKM								x	x

▼ **Driver for sizes 4 to 11**



▼ **Driver for sizes 14 to 22**



Designation	Order number EU	Order number APA / US
Driver for sizes 4 to 11	1 510 240 009	R 002 G11 025
Driver for sizes 14 to 22	1 510 240 011	F 000 511 445

Note
Spare parts can be found online at
www.boschrexroth.com/spc

Notes on commissioning

General
Pumps delivered by Bosch Rexroth are tested for function and performance. No changes of any nature may be made, otherwise the warranty is rendered void.
The pump may only be operated with the permissible data (see page 6).

Technical data
All mentioned technical data are dependent on manufacturing tolerances and are applicable for certain boundary conditions. Note that certain deviations are therefore possible and that technical data may vary when certain boundary conditions (e.g., viscosity) change.

Characteristic curves
When dimensioning the gear pump, observe the maximum possible application data on the basis of the characteristic curves shown starting on page 8.

Scope of delivery
The scope of delivery includes the components with the characteristics described under type codes and dimensions starting on page 16.

Further information
- Further information on installation, commissioning, and operation can be found in the publication 07012-B1: “General Operating Instructions for External Gear Units”.
- Extensive notes and suggestions can be found in the Hydraulic Trainer, volume 3: “Planning and Design of Hydraulic Power Systems”, order number R900018547.

Filter recommendation
Since the majority of premature failures in gear pumps occur due to contaminated hydraulic fluid, filtration should maintain a cleanliness level of 20/18/15 as defined by ISO 4406.
Cleanliness level 20/18/15 can reduce contamination to an acceptable degree in terms of particle size and concentration.

Bosch Rexroth generally recommends full-flow filtration. Basic contamination of the hydraulic fluid used may not exceed class 20/18/15 according to ISO 4406. New fluids are often above this value. In such instances, a filling device with a special filter should be used.
Bosch Rexroth is not liable for wear due to contamination.

Order number overview

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R983074477	AZPW–11– 014LCB20MB	16
R983074478	AZPW–11– 016LCB20MB	16
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R983074480	AZPW–11– 022LCB20MB	16
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R983074466	AZPW–11– 005RCB20MB	16
R983074467	AZPW–11– 008RCB20MB	16
R983074468	AZPW–11– 011RCB20MB	16
R983074469	AZPW–11– 014RCB20MB	16
R983074470	AZPW–11– 016RCB20MB	16
R983074471	AZPW–11– 019RCB20MB	16
R983074472	AZPW–11– 022RCB20MB	16
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AZ configurator

With our practical product selector, it will take you next to no time to find the right solution for your applications, no matter whether it is Standard Performance or another external gear unit.
The selector guides you through a selection of features to all of the products available for order. By clicking on the order number, you can view and download the following product information: Data sheet, dimension sheet, instruction manual, operating conditions, and tightening torques.
You can order your selection directly via our online shop and at the same time benefit from an additional discount of 2%. And if you need something really quickly, simply use our fast delivery and preferred programs (GoTo). Then the goods will be sent within 10 business days.
You also have the possibility to easily and conveniently configure your individual external gear unit with our AZ configurator. All the necessary data that you need for the project planning of external gear units is requested by means of the menu navigation.
For an already existing configuration you receive as a result the order number, the type code, as well as further information. If your configuration does not lead to a product that is available for order, our online tools provide you with the possibility of sending a project request directly to Bosch Rexroth. We will then get in contact with you.
Link: www.boschrexroth.com/az-configurator



Fit4SILENCE app

You want to quickly determine the noise level of an application but don't have a measuring device at hand? No problem with Fit4SILENCE! Our new noise measurement app for all Android devices can be immediately downloaded free of charge. After calibration, you can start using it straight away and conduct fast, accurate noise measurements with different weightings in no time at all. An additional measuring device is no longer necessary, because calibrated smartphones using the app can achieve an accuracy that approximates professional measuring devices.
Last but not least, the app contains interesting information about the SILENCE PLUS technology, including an audio sample.
Link: www.boschrexroth.com/silence-plus

▼ **Download the Android app:**



Your notes

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